



Global Supplier Quality Manual

Supplier Quality and Development

Preface

This Reference Manual represents the collaborative efforts of multiple GM worldwide teams engaged to develop a GM Common process manual that includes APQP, Launch, and Current Quality. Prior to this, multiple documents existed to define a Supplier Quality process from APQP through Current Quality.

APQP

The APQP portion of this manual defines GM's common global product quality planning requirements that are necessary to develop and implement an APQP process for a product or service. It is intended as a standard to provide the Supplier Quality Engineer, and the supplier, a common format from which to proceed with all steps of APQP.

Current

The Current portion of this manual defines the standardized work around common processes used globally to protect our manufacturing/assembly plants and to drive systemic improvement to the supply base.

Approved by:

Global Supplier Quality Directors

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Manual Content Explanation

This manual contains two (2) sections. Each section has tasks that are referenced by a task number and includes the following:

- Task name
- Task Owner(s)
- Task Timing (approximate)
- Task Description
- Key Deliverables upon completion of the specific task
- Customer(s) for the deliverables
- Necessary Inputs to complete the specific task including source for each input
- Listing of Resources—this includes people of organizations involved in completing the task
- Methodology—brief description of actions in table format arranged by action owner
- References and additional information

Document Procurement

Forms

GM forms and documents referenced in this manual GM 1927 can be obtained through GM SupplyPower at www.gmsupplypower.com and be copied for use.

Labels & Tags

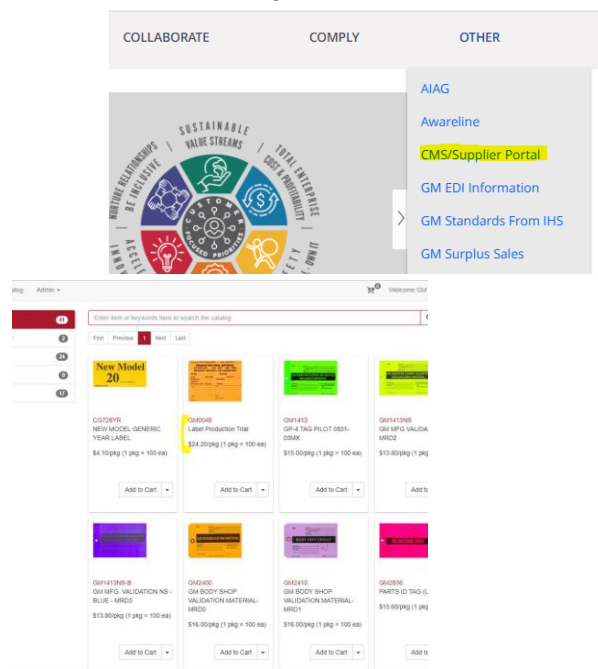
Labels and tags referenced in the APQP process can be obtained as follows:

GMAP and GME – contact the SQE

GMLAAM – Labels and tags can be obtained from any supplier as long as it meets standards identified in the General Procedures

GMNA – Contact CMS Print Solutions 248.208.0935 x 4716 ; 248.202.2744; FAX:
248.208.1672 www.cmsprintsolutions.com

- or: There is a link under the 'Other' drop-down for a 'CMS/Supplier Portal'



A supplier can then register an account and order labels.

AIAG Documents

All AIAG specific documents referenced can be obtained by contacting AIAG at 01-248-358-3003. Documents can also be ordered by accessing the web at www.AIAG.org. In Europe contact *Carwin Ltd* at 44-1708-861333.

Note to Suppliers:

This manual is intended to be comprehensive and “all-encompassing”; however, certain circumstances will prompt questions. If you have any questions regarding any part of this manual you are encouraged to contact your respective Supplier Quality Engineer

Section 1

Global APQP

Customer vs. Supplier Monitored APQP

The following matrix describes the responsibility differences between “Customer-Monitored” and “Supplier-Monitored” APQP. Suppliers are responsible for carrying out all the “R” activities shown in the supplier column of the matrix, whether or not their parts are designated as customer or supplier-monitored APQP. If a part is designated as customer-monitored APQP, a GM representative will monitor and approve the APQP activities.

APQP Activity		Customer Monitored APQP		Supplier Monitored APQP	
		Supplier	GM	Supplier	GM
1	Commodity Key Stakeholders Meeting.	N/A	R	N/A	R
2	Technical Reviews	S	R	S	R
3	Sourcing Eligibility	N/A	R	N/A	R
4	Component Readiness Valves Assessments	S R	R A	S R	R I
5	Timing Charts/Open Issues	R	A	R	I
6	GM Quality Performance Requirements (Sourceability Levels)	R	A	R	A
7	Flow Chart	R	I	R	I
8	DFMEA (1 – Supplier Design Responsible, 2 – GM Design Responsible)	R (1) S (2)	A	R S	I
9	Design Review (1 – Supplier Design Responsible, 2 – GM Design Responsible)	R (1) S (2)	S (1) R (2)	R (1) S (2)	I
10	Tool and Equipment Review	R	A	R	I
11	Gauge Review & Approval	R	A	R	A
12	PFMEA	R	I	R	I
13	Control Plan	R	I	R	I
14	Early Production Containment (GM 1927 28)	R	I	R	I
15	PPAP	R	A	R	A
16	Run @ Rate (GM 1927 35) or Build at Rate (GM 1927 35c)	R	A	R	A
17	Lessons Learned	R	I	R	I

R Responsible - Task Owner (GM or Supplier)

A Approve - Approve deliverables (GM)

S Support – Support Completion of deliverables (GM or Supplier)

I Inform - Review deliverables at discretion of SQE (GM)

N/A Not Applicable

Global APQP RASIC

APQP	Global CT SQE /GCL (multiple or one region sourcing)	DUNS /Part Owning SQE
Pre-Sourcing (APQP Tasks 1, 2, 3)		
On-time completion of Sourcing Eligibility (APQP Task 3)	R	S
SQ Participation in CKSM (APQP Task 1)	R	S
SQ support at Technical Review (APQP Task 2)	R	I
Completion of QBC QUAD Report (as required)	A	R
SQ Signature on Recommendation Sheet	R	C
Handover from Dev SQE to Mfg SQE	S	A
Post-Sourcing (APQP Tasks 4-17)		
CRV 0	S	S
CRV 1	-	S
CRV 2	-	S
CRV 3	-	S
CRV 4	-	S
CRV 5	-	R
CRV 6	-	R
PPAP (-15wks)	-	R
R@R (-8wks)	-	R
Business Processes (AUROS, SCMS, SQMS, SPPS, RLA, etc.)	-	R

Task Number: 1

Task Name: Commodity Key Stakeholders Meeting (CKSM)

Task Owner: Buyer

Task Timing: Pre-Sourcing

Task Description: Introduce and establish Key Stakeholder ownership, identify program-specific strategies in the Engineering & Advance Purchasing Sourcing Process (E&APSP) and ensure that the RFQ-package contains all information needed to receive comparable quotes.

Key Deliverables:

- SQ approval of the RFQ package content, timing and preliminary Sourceability Report

Customer for Deliverables: Purchasing, Engineering, Supplier Quality and Marketing, PC&L as required

Necessary Inputs:

Source of Input:

- | | |
|---|--------------------|
| • Technical documents (BOM, SOR & Appendices e.g. SSTS/CTS) | Engineer |
| • Creativity Team Bidders List (enhanced quality metrics) | Buyer |
| • Proposed Sourcing Strategy | Buyer |
| • Part-Specific Quality & Process CG | SQE |
| • Lessons Learned (including warranty items) on previous programs | Buyer/SQE/Engineer |
| • Program related information | Key Stakeholders |

Resources: Purchasing, Supplier Quality, Engineering

<u>Responsible</u>	<u>Methodology :</u>
Buyer	Invite appropriate Key Stakeholders identified for specific component/commodity/system and conduct the meeting as stated on typical Engineering & Advance Purchasing Sourcing process (E&APSP) agenda (this meeting is to be conducted on all parts listed in the sourcing plan at the time stated in the sourcing plan).
Buyer	Clarify the timing of each step in the AP Sourcing process, what is expected from each Stakeholder and get agreement on the process, content, timing and strategy for the sourcing package.
Buyer	Establish timing and lead question development for Supplier Workshops (Technology, Cost Reduction or Problem Resolution). Define a core group (sub-group of Stakeholders) to attend Supplier Workshops, summarize results and refine functional requirements based on workshop findings (if applicable).
Buyer	Apply E&APSP CKSM & SOR checklists as appropriate.
Buyer	Start Open Issues list with any items that need to be addressed.
DRE	Present SOR content & allowable cost status and clarify what is being sourced (technological, dimensional, functional and test requirements).
DRE	Present Warranty Data – i.e. Incidents per Thousand Vehicles (IPTV) , Cost per Vehicle (CPV) and engineering benchmark data (consult warranty champion if necessary).
SQE	Review potential bidders list proposed by Buyer to restrict Request for Quotation (RFQ) distribution to suppliers that meet GM criteria. Identify suppliers that will require Quality Business Case action plans as established by GM Global APQP task #3 and start planning any required audits.
SQE	Review the requirements stated in the CG4338 GM 1927 03 Supplier Quality Statement of Requirements, CG4355 GM 1927 03 Sub Tier Supplier Management and the Part-Specific Quality & Process CG (if applicable) to ensure that they are included in the Request For Quotation package.
SQE	Provide an overview of commodity specific Lessons Learned from previous programs.
SQE	Inquire about any sequencing plans.
SQE	Apply the Commodity Key Stakeholders Meeting Checklist GM 1927 06 as appropriate.

Reference Documents:

- CG4338 GM 1927 03 Supplier Quality Statement of Requirements
- CG4355 GM 1927 03 Sub Tier Supplier Management
- CGxxxx GM 1927 03 Part-Specific Quality & Process
- GM 1927 05 APQP Open Issues list
- GM 1927 06 Commodity Key Stakeholders Meeting Checklist

Task Number: 2

Task Name: Technical Review

Task Owner: Buyer

Task Timing: Pre-Sourcing

Task Description: Review the supplier technical proposal to ensure that all requirements in the RFQ-package have been understood and supplier has a plan to produce parts meeting GM expectations.

Key Deliverables:

- Identification of suppliers that are capable
- Completed GM 1927 07 APQP Supplier Assessment

Customer for Deliverables: Buyer, SQE, Engineer, Manufacturing Engineer

Necessary Inputs:

- Technical documents
- Supplier's Quality History
- List of Directed Buys
- Carryover part contact list
- Required Quality Information as per CG4338 GM 1927 03 SQ SOR

Source of Input:

Engineer
SQE
Buyer
Buyer
Supplier

Resources: Purchasing, Engineering, Supplier Quality, Manufacturing Engineer, Supplier

<u>Responsible</u>	<u>Methodology :</u>
Buyer	Invite potential suppliers to Technical Review Meetings.
Buyer	Update the Open Issues List with concerns related to the supplier's ability to meet the requirements.
DRE	Identify specific agenda items for Suppliers to cover in Technical Review.
DRE	Assess Supplier technical capability for continuation in the sourcing process.
SQE	At the conclusion of the Technical Review, the Buyer, SQE and Engineer complete the GM 1927 07 APQP Supplier Assessment (this is required prior to signing the Sourcing Recommendation form).
SQE	Prior to the meeting, review specific documents provided in the supplier's quote package as per CG4338 GM 1927 03 SQ SOR and prepare questions related to quality and manufacturability in order to assess supplier capability and to support SQ recommendation about supplier continuation in the sourcing process.
SQE	Ensure Supplier understands the CG4338 GM 1927 03 SQ SOR, and Part-Specific Quality & Process CG (if applicable) and returns them properly signed.
SQE	Provide a brief overview of the Lessons Learned process.
SQE	Complete the GM 1927 13 Technical Review Checklist (required)
Supplier	Review info as requested by "Required Quality Information" item in the CG4338 GM 1927 03 SQ SOR: Preliminary timing charts: • Highlight concerns relative to tooling, gauges or testing that may impact PPAP • Revisions to manufacturing facility, including Greenfield or Brownfield plans • Manpower resource commitment to ensure successful completion of program; proper skills and training to perform the necessary tasks. Preliminary process flow diagrams, PFMEA and Control Plan: • Highlight any special assembly techniques, test methods or containment procedures used. Capability studies on similar parts : • Confirm error proofing, data analysis, and record keeping is included and review plan to meet any GM warranty targets (review existing open SPPS records and warranty sufficiency/action plans, data on similar products and plans for future reductions). Tiered supplier management: • Present process and resource plan for evaluation and management of tiered suppliers.

Supplier	Provide evidence of prior product experience or technical expertise relevant to new product.
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Reference Documents:

- CG4338 GM 1927 03 Supplier Quality Statement of Requirements
- CGxxxx GM 1927 03a Part-Specific Quality & Process
- GM 1927 05 APQP Open Issues list
- GM 1927 07 APQP Supplier Assessment
- GM 1927 13 Tech Review Checklist

Task Number: 3

Task Name: Sourcing Eligibility

Task Owner: SQE

Task Timing: Pre-sourcing

Task Description: For a supplier location to be eligible to receive a new business award from GM the manufacturing location to be sourced must either be:

- Green on the GM Creativity Team Sourceability Report for Quality for the specific commodity, OR
- Have a supporting Quality Business Case (QBC) QUAD report approved according to the criteria outlined in this task.

Key Deliverables:

- SQ Signed GPS recommendation sheet or equivalent and appropriate back-up documentation – either a copy of the latest GM Creativity Team Sourceability Report OR an approved QBC QUAD report with supporting audit approval

Customer for Deliverables: Purchasing

Necessary Inputs:

- Bidding DUNs number / Supplier manufacturing location list
- Minimum information by location:
 - Supplier name, address and DUNs number (if available)
 - Greenfield/Brownfield Y/N,
 - Existing IATF 16949 certification Y/N,
 - Location currently ships specific commodity to GM Y/N
- Assessment required completion date
- Supplier contact details by location
- Current GM Creativity Team Sourceability Report report Quality status (R/G/NR)

Source of Input:

Buyer
Buyer
Buyer/SQE
Buyer/SQE
Buyer/SQE
Buyer
Buyer
Buyer/SQE

Resources: Purchasing, Supplier Quality and Supplier.

<u>Responsible</u>	<u>Methodology :</u>
Buyer	Provide final supplier location inputs, required to determine audit requirements, a minimum of 2 weeks prior to nomination date (information should be provided pre-Technical Review where feasible).
SQE	Based on the input information provided, using the table below, determine any audit required prior to sourcing for each supplier location in order to assess sourcing eligibility: • Where supplier location has IATF 16949 certification the audit requirement is PCPA, otherwise PSA audit should be applied.
SQE	Work jointly with the Sourcing Team to identify any DUNs numbers not Green for Quality on the GM Creativity Team Sourceability Report at time of nomination and plan audits / QBC QUAD report follow-up accordingly.
SQE	Work with the Regional SQE (if different) to ensure audit scheduling and completion ahead of nomination date as required.
SQE	Work with the Regional SQE (if different) to ensure signed QBC QUAD report is provided ahead of the nomination date as required.
SQE	Communicate regularly with Buyer to comprehend any supplier location changes and, in particular, communicate any rejections / non-sourceable supplier locations as early as possible for escalation to CT.
SQE	Ensure completion of required audits in time for nomination and provide signed QBC QUAD report to Buyer where necessary ahead of sourcing table.
SQE	For a new supplier location sourcing: In QBC QUAD report, capture all open action items and plans for correction/mitigation and obtain SQ approval (from region where the supplier is located) to support the sourcing recommendation.
SQE	Sign sourcing recommendation for selected supplier.
Supplier	Provide necessary information as required and support audit requests at short notice.

Task Name: Sourcing Eligibility**Sourcing Approval Matrix:**

Supplier Location Type	Sourceability Report Quality Rating	IATF 16949 Quality Standard Certified?	Required Audit	Comments
Greenfield (New Construction) OR Brownfield (Existing Building)	N/A	No, but Sister Facility Yes.	PCPA on Sister Facility Building Like Product.	Company Performance Data from Other Customers or Internal Data. QBC from Regional SQ Required. Buyer Must Request Duns be Loaded into CSIDS when Available.
Greenfield (New Construction) OR Brownfield (Existing Building)	N/A	No, and Sister Facility No.	PSA on Sister Facility Building Like Product.	Company Performance Data from Other Customers or Internal Data. QBC from Regional SQ Required. Buyer Must Request Duns be Loaded into CSIDS when Available.
Current Supplier to GM	Green	Yes	None	SQ signature / approval required on GPS Recommendation Sheet or Equivalent.
Current Supplier to GM	Red	Yes	None	Location cannot be sourced. Refer to SQ Regional Director.
Current Supplier To GM New CT Different Product	Green	Yes	PCPA	Acceptable PCPA, QBC (Quad Report) from Regional SQ Required. SQ signature / approval required on GPS Recommendation Sheet or Equivalent.
Current Supplier to GM New CT Different Product	Red	Yes	None	Location cannot be sourced. Refer to SQ Regional Director.
Current Supplier to GM New CT Different Product	NR	Yes	PCPA	Acceptable PCPA. Need Company Performance Data from Other Customers or Internal Data. QBC from Regional SQ Required. SQ signature / approval required on GPS Recommendation Sheet or Equivalent.
New to GM, no business with any GM locations	NR	Yes	PCPA	Need Company Performance Data from other Customers or Supplier Internal Data. QBC from Regional SQ Required.
New to GM, no business with any GM locations	NR	No	PSA	Need Company Performance Data from other Customers or Supplier Internal Data. QBC from Regional SQ Required.
New DUNS #, other Duns currently ship to GM	NR	Yes	PCPA	Review data from other Duns locations. QBC from Regional SQ Required.
New DUNS #, other Duns currently ship to GM	NR	No	PSA	Review data from other Duns locations. QBC from Regional SQ Required.
Not on Sourceability Report. Currently Shipping to location that is not using SCMS	N/A	Yes	None	Current Local Sourceability Report Showing Green on Quality Metrics or Local SQ Approval. – Only Valid for Additional Business to Current GM Customer Location.
Not on Sourceability Report. Currently Shipping to location that is not using SCMS	N/A	Yes	PCPA	Current Local Sourceability Report Showing Green on Quality Metrics. QBC from Regional SQ Required. – Only Valid for Additional Business to New GM Location.
Not on Sourceability Report. Currently Shipping to location that is not using SCMS	N/A	No	PSA	Current Local Sourceability Report Showing Green on Quality Metrics. QBC from Regional SQ Required. – Only Valid for Additional Business to New GM Location.
No Duns Number	N/A	Yes	PCPA	Apply to Dun and Bradstreet for Duns Number. Follow Quality, Audit and Sourceability Report Requirements Previously Stated Depending on Supplier Situation. Buyer Must Request Duns be Loaded into CSIDS when Available. QBC from Regional SQ Required.
No Duns Number	N/A	No	PSA	Apply to Dun and Bradstreet for Duns Number. Follow Quality, Audit and Sourceability Report Requirements Previously Stated Depending on Supplier Situation. Buyer Must Request Duns be Loaded into CSIDS when Available. QBC from Regional SQ Required.

Reference Documents:

- GM 1927 08 Potential Supplier Assessment (PSA) audit
- GM 1927 16 Process Control Plan Audit (PCPA)
- GM 1927 71 QUAD report form

Task Number: 4

Task Name: APQP Kick Off meeting and Component readiness valves

Task Owner: See below:

4

APQP Kick Off meeting (Optional): SQE

Component readiness valves:

CRV	0	1	2	3	4	5	6
CRV Leader	AVD DESIGN RELEASE ENGINEER	DESIGN RELEASE ENGINEER	DESIGN RELEASE ENGINEER	BUYER	VALIDATION	SUPPLIER QUALITY ENGINEER	SUPPLIER QUALITY ENGINEER
CRV Participants	Cross-functional PDT	Cross-functional PDT	Cross-functional PDT	Cross-functional PDT	Cross-functional PDT	Cross-functional PDT	Cross-functional PDT

Task Timing: Follow the CRV timeline.

Task Description: APQP Kick Off meeting and Component Readiness Valve (CRV) are a cross-functional assessment of PDT standard work. CRV Assessments proactively confirm PDT Quality Chain standard work is completed at key milestones in the development process. Intent is to enable defect free product and process designs across GVDP/GPDP by assessing Quality Chain deliverables at various milestones.

Key Deliverables:

- Completed CRV 1,2,3,4,5,6 and record in Auros. (Completed CRV 0 when applicable).

Customer for Deliverables: SQE, Buyer, Engineer, Launch Leader, Manufacturing as appropriate.

Necessary Inputs:

- Program timing for key events
- Detailed program timing
- GM 1927 05 APQP Open Issues List (optional)

Source of Input:

Engineer/Buyer
Supplier
Supplier/SQE

Resources: Buyer, Engineer, SVE, SQE, Manufacturing Engineer

Task Name: APQP Kick Off meeting and Component readiness valves

Responsible	Methodology:
SQE	SQE conducts the Kick Off meeting offline (optional).
SQE	Supporting CRV 0-CRV 4 where applicable
SQE	Leading CRV 5- CRV 6
Supplier	It is the responsibility of the Supplier to ensure that the process meets or exceeds all requirements and able to show compliance through the associated Commodity Specific Audit (self-audit and/or through onsite audit by GM personnel) within 30 days from sourcing and verified according to CRV process.

Reference Documents:

- GM 1927 05 APQP Open Issues List
- GM 1927 11 APQP Kick Off meeting
- GM 1927 16a Commodity Audit
- CG4338 GM 1927 03 Supplier Quality Statement of Requirements
- CGxxxx GM 1927 03a Part-Specific Quality & Process CG

Task Number: 5

Task Name: Timing Chart and APQP Open Issues List

Task Owner: Supplier

Task Timing: Reviewed at CRV Assessments and throughout the program

Task Description: A detailed review of all timing charts and concerns is conducted at least every 2 months to ensure that program deliverables are executed on schedule. These reviews are conducted on GM monitored parts tracked using the APQP process. For supplier monitored APQP, the supplier reviews and updates the Timing Chart and Open Issues List throughout the APQP process.

Key Deliverables:

- Up-to-date GM 1927 02 APQP Timing Chart or equivalent format containing all required information.
- GM 1927 05 APQP Open Issues list

Customer for Deliverables: Supplier Quality Engineer

Necessary Inputs:

- Open issues
- Detailed timing for tooling, facilities, gauges, etc.

Source of Input:

SQE/Supplier/Engineer/Buyer
Supplier

Resources: Supplier, SQE, Engineer, Manufacturing Engineer, and other members of the program team as appropriate

<u>Responsible</u>	<u>Methodology:</u>
SQE	Review the APQP Timing Chart and APQP Open Issues List, minimum monthly.
SQE	Drive supplier to develop recovery plans on issues impacting timing, quality or capacity.
SQE	Identify and communicate key timing and program issues to GM management.
Supplier	Create an GM 1927 02 APQP Timing Chart or equivalent and track tasks to completion in line with chart. Create an GM 1927 05 APQP Open Issues List and present updates to SQE at all Component Readiness Valve (CRV) Assessments.
Supplier	Sub-component milestones shall be pulled ahead of Tier 1 assembly milestones by 6 weeks to protect GM program timing.
Supplier	Update timing chart as timing changes occur and communicate any changes, concerns and issues to SQE on an ongoing-basis (maintain additional detail behind each high level APQP Timing Chart item, additional detail must be tied to high level chart to ensure timing is updated automatically).
Supplier	Develop recovery plans for issues impacting timing and drive the plan to maintain program timing.
Supplier	Utilize APQP Open Issues list to capture all issues requiring action.

Reference Documents:

- GM 1927 02 APQP Timing Chart
- GM 1927 05 APQP Open Issues list

Task Number: 6

Task Name: GM Quality Performance Requirements (QPR)

Task Owner: Supplier

Task Timing: APQP Kick-off (or within 30 days from sourcing)

Task Description: Suppliers are required to obtain and maintain compliance to GM QPR which include the following criteria:

- 1) IATF 16949 Certification
- 2) Number of Field Actions
- 3) Number of Major Disruptions
- 4) Number of Quality SPPS in 6/12 months
- 5) Severity Score for 6/12 months

Key Deliverables:

- If the Supplier is not IATF 16949 Certified - Action plan for IATF 16949 certification and/ or GM 1927 31 Greenfield/Brownfield Development Assessment.
- If the supplier is IATF 16949 Certified and not meeting GM Quality Performance Requirements they must submit to GM SQE an Action Plan to address the deficiencies. The Action Plan must be approved by GM SQE.

Customer for Deliverables: Supplier Quality

Necessary Inputs:

- IATF 16949 Certification
- Sourceability / Impact detail report

Source of Input

- IATF data base
- SCMS

Resources: SQE and Supplier

<u>Responsible</u>	<u>Methodology:</u>
Supplier	The supplier needs to demonstrate that they have a tool to assess their quality system and compliance to Quality Performance Requirements. The supplier's quality system assessment should be to further develop sustainable quality systems and for Continuous Improvement.
SQE / Buyer	For Greenfield suppliers, apply the GM 1927 31 Greenfield & Brownfield Development Assessment.
Supplier	If the supplier is not meeting GM Quality Performance Requirements, then supplier must create an Action Plan to meet the requirements. The Action Plans must be completed with appropriate detail, timing, ownership, etc. The Actions Plan shall be approved by GM SQE.
SQE	Monitor the Supplier Action Plan on a timely basis.
SQE	Failure to implement all Action Plan's items, could result in the SQE issuing a <i>Program Management SPPS</i> .
Supplier	Implement and verify effectiveness of all the action plan items before the next CRV.
Supplier	Keep the customer updated regarding implementation status and any issues or concerns with implementation.

Reference Documents / Tools:

- IATF 16949 Requirements
- GM 1927 30 Quality Management Gap assessment (recommended)
- GM 1927 31 Greenfield/Brownfield Development Assessment
- GM 1927 36 Quality Management (QM) Elements Training (5 modules in SupplyPower) (recommended)
- GM 1927 78 Step Down Chart

Task Number: 7

Task Name: Process Flow Charts

Task Owner: Supplier

Task Timing: Initial chart at Technical Review, **reviewed at CRV Assessments**

Task Description: Flow chart provides a logical pictorial representation of the process flow that can be used as the foundation for PFMEA's, control plans, work station layouts, etc.

Key Deliverables:

- Process Flow Chart
- Process Flow Chart depicting sub component production system

Customer for Deliverables: Supplier Quality Engineer

Necessary Inputs:

- Information on each step of the manufacturing process
- Part-Specific Quality & Process CG (if applicable)

Source of Input:

Supplier
SQE

Resources: Supplier, SQE, Manufacturing Engineer

<u>Responsible</u>	<u>Methodology:</u>
SQE	Review the preliminary process flow chart prior to sourcing to ensure completeness.
SQE	Review the production flow chart for completeness and continue with a comparison to the production line: • Ensure the flow chart is linked to the PFMEA and control plan. • Walk the manufacturing line to ensure the chart is representative of the process and includes receiving, storage, production, inspection, rework, packaging, labeling and shipping. • Ensure Part-Specific Quality & Process CG (if applicable) is applied to support flow chart development. • Ensure completion in line with AIAG A-6
Supplier	Create a preliminary process flow chart using a similar process as part of bid package requirement.
Supplier	Define a production flow chart once product design is released.
Supplier	Document all items in the flow chart with the respective nomenclature (store, move, inspect, correct, etc.)
Supplier	Ensure the process flow chart is linked to the PFMEA and control plan and complete AIAG A-6.
Supplier	Update flow chart to reflect actual production process.
Supplier	Communicate any changes on an ongoing-basis to SQE.

Reference Documents:

- AIAG Advanced Product Quality Planning and Control Plan manual Process Flow Chart checklist A-6
- AIAG PFMEA 4th Edition or AIAG FMEA VDA Handbook and CG4338 GM 1927 03 SQ SOR.

7

Task Number: 8

Task Name: DFMEA

Task Owner: Design Owner (GM Engineer or Supplier)

Task Timing: Initiated before or at design concept

8

Task Description: The DFMEA is a living document that is initiated before or at design concept and is continually updated as changes occur, or additional information is obtained throughout the phases of product development. It supports the design process in reducing the risk of failure by: 1) aiding in the evaluation of design requirements, DFM, and DFA, 2) increasing the probability that potential failure modes have been considered and 3) establishing a priority system for design improvements. 4) Any Potential Failure modes that have a sensitivity to the Process capability need to be communicated and referenced in the PFMEA. 5) Special Care Characteristics are identified in the DFMEA Analysis and are referenced to the Potential Failure modes. 6) Lessons learned are updated and identified in the DFMEA.

Key Deliverables:

- DFMEA /DRBFM

Customer for Deliverables: Engineer, Supplier, Supplier Quality Engineer

Necessary Inputs:

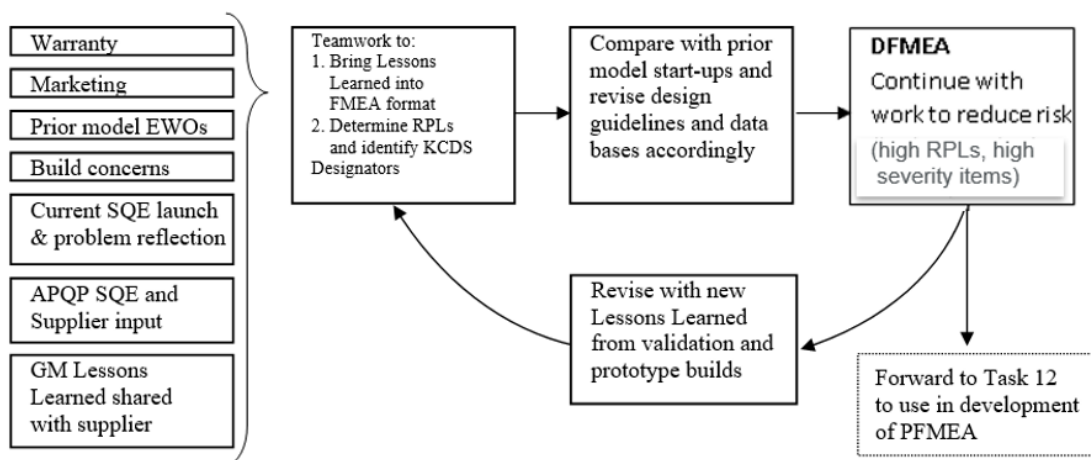
- Requirements as stated in SOR/SSTS/CTS/CG
- Design intent
- Vehicle requirements
- Manufacturing/Assembly requirements
- Lessons Learned /Part-Specific Quality & Process CG
- KCDS Templates (if already identified)

Source of Input:

Engineer
Design Responsible Engineer
Engineer
Design Responsible Engineer
SQE/Engineer
Engineer

Resources: Engineer, Design Responsible Engineer

Methodology:



Responsible	Methodology
SQE	Confirm that the Supplier (if not design responsible) is working with the GM Engineer on development of the DFMEA, or DRBFM for any design changes made after production release. If not, take appropriate action to initiate this team activity.

SQE	Confirm with Design Owner if the DFMEA is updated with the results of the DRBFM analysis.
SQE	Confirm a DFMEA has been completed by the responsible Design Owner and confirm that the supplier has received DFMEA Severity 9 and 10's and associated Potential Failure Modes. Confirm that the supplier has access to necessary information from the GM DFMEA as input into the PFMEA.
SQE	Work with supplier to confirm that a DFMEA and System FMEA have been completed for all sub components by the responsible engineering function or the sub-contractor.
Design Owner	Lead the workshops with GM DRE, SQE and Supplier Engineering and complete the DFMEA. In addition, complete the Design FMEA checklist (AIAG A-1). Develop and implement RPL reduction plans and strive to continuously reduce RPL, provide a DRBFM analysis for any design changes made after production release, and update the DFMEA with the results of DRBFM analysis.
Supplier	If not design responsible, provide any lessons learned to the GM Engineer and support the development of the DFMEA, and the DRBFM (if applicable). In addition, complete the Design FMEA checklist (AIAG A-1).
Supplier	If GM is design responsible and does not provide access to necessary information from the DFMEA or DRBFM (if applicable), document this issue on the GM 1927 05 APQP Open Issues List.
Supplier	If the subcontractor is design responsible ensure that the DFMEA is complete for all sub-components. Ensure all Severity 9&10's have been communicated to the Subcomponents. Monitor and drive the development and implementation of RPL reduction plans on sub components.
Supplier	If the subcontractor is not design responsible ensure that any lessons learned are provided to GM engineering for input in developing the DFMEA.

Reference Documents:

- GM 1927 05 APQP Open Issues list
- AIAG Advanced Product Quality Planning and Control Plan manual Design FMEA checklist A-1
- KCDS Manual GMW15049
- Generic Supplier Analysis/Development/Validation Tasks & Deliverables for All Programs & Commodities GMN 3600
- Design Review Based on Failure Modes GMN 11220

Task Number: 9

Task Name: Design Reviews

Task Owner: Design Owner (GM Engineer or Supplier)

Task Timing: Initial review–prior to CVER, SVER & IVER; subsequent reviews occur on an on-going basis

Task Description: Reviews to ensure the design has been adequately defined to enable construction of tools and gauges.

Key Deliverables:

- Defined and measurable KCDS Designators
- GD&T
- Appearance, performance and material testing specifications

Customer for Deliverables: Supplier

Necessary Inputs:

- SOR / SSTs / CTS /CG
- Appearance specifications
- Performance and material specifications
- Production Assembly Documents if available
- Bill of Material (BOM)

Source of Input:

Engineer
Engineer
Engineer
Engineer
Engineer

Resources: Engineer, Supplier Quality Engineer, Supplier, Subcontractors

<u>Responsible</u>	<u>Methodology:</u>
Design Owner	Schedule and conduct KCDS workshop. SQE, GM Engineer and supplier participate in workshop. The purpose of the workshop is to define critical product characteristics, and critical process control characteristics.
Design Owner	Schedule and conduct GD&T review. The SQE, GM Engineer and supplier participate in the review. The purpose of the review is to define design dimensioning and tolerance on the drawing as these items relate to the actual function of the part.
Design Owner	For subcomponents, the same process is used to determine lead responsibility in conducting these reviews. Supplier and SQE participate as necessary in the reviews
DRE	GM Engineer is responsible to ensure that all information available in the KCDS template at KCDS website for a specific commodity has been applied, even when supplier is design responsible.
SQE	Ensure the GD&T datum scheme guarantees proper gauging and that there are GD&T callouts for all features which are basic dimensions (performed by SQ Gauge Engineer, if available in the region).
SQE	Confirm the selected KCDS Designators can be measured with variable or attribute gauges.
SQE/Supplier	Confirm the manufacturing process can achieve the specified tolerances on a sustained basis.
Supplier	Any recommendations / improvements to the design record are documented and submitted to GM for approval.
Supplier	Communicate any concerns relative to the KCDS Designators or GD&T scheme and the manufacturability of the part. Document any open issues on the APQP Open Issues List GM 1927 05.
Supplier	Develop an understanding of the engineering change process with the GM Engineer. Apply AIAG Checklist (A-2) and address any open action item.

Reference Documents:

- GM 1927 05 APQP Open Issues
- GMW15049 KCDS Manual
- AIAG Advanced Product Quality Planning and Control Plan manual
- AIAG Design Information Checklist A-2
- GMW3059 General Motors Global Standard, Restricted and Reportable Substances for Parts
- GMW3116 Recyclability/Recoverability Design Guide

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Task Number: 10

Task Name: Tooling and Equipment Reviews

Task Owner: Supplier

Task Timing: From Tooling Kick Off Meeting (TKO) through PPAP approval.

Task Description: Ensure that the manufacturing process is being designed to the latest drawing change level, built and certified to produce parts with quality at rate according to GM program requirements.

Key Deliverables:

- Process capable Tooling and Equipment delivered on time.

Customer for Deliverables: Supplier Quality Engineer, Engineer

Necessary Inputs:

- Approved GD&T, KCDS Designators and Math Data
- Part-Specific Quality & Process CG (if applicable)
- Process Capability Requirements
- Timing requirements
- Capacity requirements
- Error proofing
- Functional Test Requirements

Source of Input:

Engineer
SQE
SQE / Supplier
Buyer / SQE / Supplier
Buyer
Supplier
Engineer/Supplier

Resources: Engineer, Supplier, Supplier Quality Engineer, Manufacturing Engineer, DRE

<u>Responsible</u>	<u>Methodology:</u>
SQE	Starting at Tooling Kick Off meeting review tooling and equipment concept and pass / fail criteria for component functional test to ensure they meet Part-Specific Quality & Process CG, they can meet product design intent and achieve process capability requirements and ensure supplier has submitted a detailed timing chart for tooling and equipment development. Conduct equipment reviews throughout build process.
SQE	Review timing to obtain tools and equipment and functional test in line with program targets for, PPAP and Run @ Rate. Verify that the supplier has planned (at minimum) and is working to complete and / or tracking the following activities at appropriate time: • GM Engineering Design Releases • Manufacturability Assessment and Tooling Concept • Tooling Design • GM Tooling Purchasing order • Tooling Construction & Try-Out at Tool Shop • Tooling Buy-Off with final equipment at final production location previously to PPAP run Conduct tooling reviews & audits throughout build process.
DRE/SQE	Review timing for Pull Ahead Production Tools with supplier to ensure compliance to IVER/PVVB/Gamma build requirements.
SQE	At the tooling buy-off or PPAP approval, verify the existence of GM owned tooling and that it is properly identified as GM property.
Supplier	Review all timing and inform GM SQE about any changes from original project timing.
Supplier	Design, build and obtain certification of and validation of tooling and equipment as per latest math data.
Supplier	Participate in design reviews and ensure GD&T, KPC, PQC and AQC requirements are included in the tooling, equipment and functional test design. Also ensure tooling and equipment will produce parts to process capability requirements. GM 1927 20 Process Capability Measurement System Analysis (MSA)
Supplier	Inform GM about any design and process changes regarding any new or modified tools, rearrangement of existing tooling or equipment, any product or process changes impacting fit, form, function, performance and durability of saleable product, any test/inspection methods and any new facilities.
Supplier	Prior to tools being grained, dimensional verification is required by GM and Approval to Grain form signed (verify with SQE on regional requirements).
Supplier	Ensure PFMEA has been comprehended and incorporated into tooling and equipment design.
Supplier	Complete the AIAG APQP New Equipment Checklist (A-3).

Reference Documents:

- AIAG Checklist A-3
- GM 1927 20 Process Capability Measurement System Analysis (MSA) Result Sheet

Task Number: 11

Task Name: Gauge Development & Approval

Task Owner: Supplier

Task Timing: Starting (Concept) at P3 release and complete by PPAP Approval

Task Description: To design, build and certify gauges according to latest GD&T release and per GM requirements.

Key Deliverables:

- GM 1927 29 Gauge Concept Design and Build review workbook fully approved by SQE or GM Gauge Engineer 8 weeks prior to PPAP date

Customer for Deliverables: GM Gauge Engineer, GM Supplier Quality Engineer, Manufacturing Location

Necessary Inputs:

- Part Math Data (P2) and GD&T (P3=P2 + 4 weeks)
- KCDS Designators
- Gauge Concept, Design and Build Approval
- Timing requirements

Source of Input:

Dimensional Engineer and Design Engineer
Design Release Engineer
GM Gauge Engineer
Buyer / SQE / Supplier

Resources: Supplier, GM Gauge Group, SQE, GM Design Release Engineer, Dimensional Engineering

Responsible	Methodology:
SQE/SQGE If required	Starting at APQP Kick-off meeting (or within 30 days of sourcing), ensure supplier is aware of GM 1927 29 Gauge Concept Design and Build review workbook and GM 1927 10 Fixture Standards and ensure supplier understands the gauge requirements. Supplier should have a detailed timing chart for gauge development.
SQE / SQ Gauge Engineer (SQGE), if required	Review plans and timing to obtain gauges in line with program targets for parts, PPAP and Run @ Rate. Ensure supplier is tracking and executing key tasks on time as Purchase orders issue, Gauge Concept, Gauge Design, Construction, Dimensional Certification and GR&R Validation. Conduct gauge build reviews at Gauge Shops or supplier location
SQE / SQGE if required	Concept Review -Review GD&T, part data and gauge expectations starting at Concept Review meeting. Include the GM Dimensional Eng.(DSE) and the GM Design Release Eng. (DRE)
SQE / SQGE if required	Design Review -Review completed gauge design(s) based on gauge concepts per the GM 1927 29. . Include the GM Dimensional Eng.(DSE) and the GM Design Release Eng. (DRE)
SQE / SQGE If Required	Build Review - Review construction to ensure that gauge meets latest GD&T, agrees with functional part usage / datum strategy and includes measurement of KCDS designators if required, per the GM 1927 29. Include the GM Design Release Eng. (DRE)
SQE / SQGE If required	Approve gauge Build Review per GM 1927 10 GM Fixture Standards, including: • Ensure that any part changes are incorporated into gauges. • Evaluate Coordinate Measuring Machine (CMM) report to ensure gauge accuracy. • Ensure gauge instructions (ODS) are available at the manufacturing operation. • Verify integrity of gauge for fit and function and GR&R (reference MSA AIAG). • Verify that it is properly identified as GM property.
Supplier	Deliver GM 1927 29 Gauge Concept, Design and Build Approval Workbook at APQP Kick-off meeting (or within 30 days of sourcing), ensure the gauge will be ready when the first part comes off the tool, and add to PPAP package once gauge is finished, form has all tabs filled out and it is approved by GM SQE or GM SQ Gauge Engineer, if available in the region.
Supplier	Review timing to ensure compliance and inform GM SQ Gauge Eng. and SQE about any changes from original project timing, any design and process changes, any new part math data and/or GD&T changes, any test/inspection methods that include gauges, any new facilities that would house the gauges.
Supplier	Participate in concept, design and build reviews. Ensure GD&T strategy/drawing is released and variable data collectors required by KPC, PQC, DR and AQC are incorporated into final gauge design if required.
Supplier	Design, build and certify gauge dimensionally (including a third party certification, when required), per form complete AIAG MSA GR&R (and Bias study, if required) and meet all requirements established

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	by GM 1927 10 GM Fixture Standards. A lean gauge (pull ahead CMM holding fixture) should be available for first IVER build.
Supplier	Deliver AIAG MSA studies (BIAS, Linearity, Stability, GR&R and Fleiss Kappa, if applicable) for all measurement devices and add to the PPAP package. Reference form GM 1927 20 Process Capability Measurement System Analysis (MSA) Result Sheet or similar.

Reference Documents:

- AIAG Measurement Systems Analysis Manual (Latest Edition)
- GM 1927 10 General Motors Fixture Standards
- GM 1927 29 Gauge Concept, Design and Built Approval Workbook
- GM 1927 20 Process Capability Measurement System Analysis (MSA) Result Sheet

Task Number: 12

Task Name: PFMEA

Task Owner: Supplier

Task Timing: Starting with preliminary PFMEA evaluation at Technical Review. Complete by PPAP

Task Description: The PFMEA is a living document that is initiated before or at Design Concept and is continually updated as changes occur, or additional information is obtained throughout the phases of process development. Ensure that all the requirements (internal and external) are clearly identified and as a consequence all potential failure modes of the process have been considered and addressed to reduce risk of defects through Risk Limiting Method (RLM)

Key Deliverables:

- PFMEA
- GM 1927 21 DFMEA / PFMEA Gap Analysis and Risk Mitigation Action Plan
- GM 1927 21 (to be uploaded in SQMS under the part number, under Process Documentation Element)

Customer for Deliverables: Supplier Quality, Supplier

Necessary Inputs:

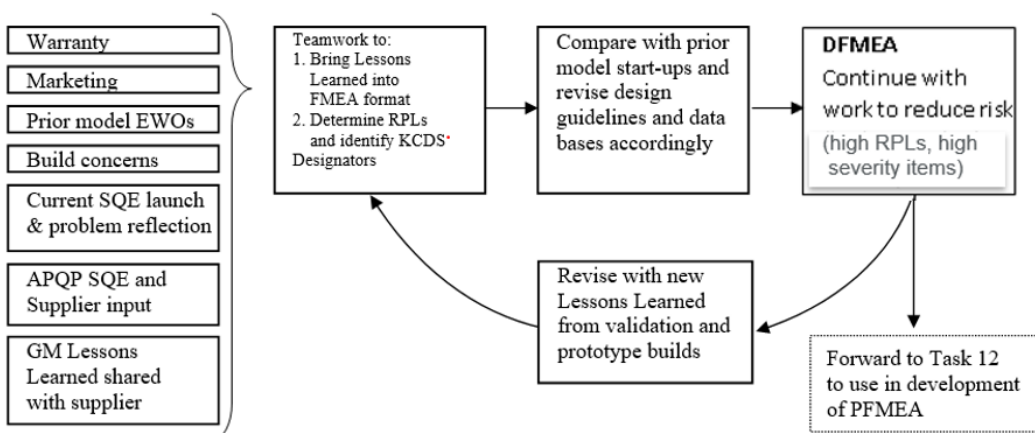
- DFMEA
- Lessons learned and Part-Specific Quality & Process CG
- Process Flow chart
- Warranty data
- SPPS records on similar parts
- Supplier Performance Report
- Supplier's manufacturing process capability on similar parts
- Error proofing techniques
- DFM / DFA Workshop results

Source of Input:

Engineer/Supplier
Supplier/SQE
Supplier
Engineer/SQE
Supplier/SQE
Supplier/SQE
Supplier
Supplier/SQE
Engineer/SQE/Mfg

Resources: Supplier Quality, Engineering, Supplier, Manufacturing Engineer and *Assembly Plant

Methodology:



Responsible	Methodology
SQE	Attend initial PFMEA development team meeting, provide GM data (Warranty, SPPS records, Lessons learned, etc.), and discuss PFMEA methodology.
SQE	Monitor progression of PFMEA development and confirm participation of multiple cross-functional team members.
SQE	Ensure Part-Specific Quality & Process CG (if applicable) is applied to support PFMEA development.
SQE	Apply the GM 1927 37 PFMEA Audit and review GM 1927 21 DFMEA PFMEA Gap Analysis Process and Transition Form (RPL Reduction summary plans), and monitor progress at each Valve Assessments and ensure that action plans have been adequately defined for high RPLs and high severity failure modes as required.
SQE	Ensure supplier has completed PFMEA reviews with subcontractors.
Supplier	Initiate PFMEA prior to sourcing as part of the bid package. This preliminary PFMEA should include critical error prevention and error detection ideas and consider any lessons learned from previous programs.
Supplier	Review PFMEA and update it as necessary each time a design change is made or a processing change is implemented. Any and all potential areas for failure are included in the PFMEA and appropriate corrective actions implemented (a PFMEA should not consider product design changes to compensate the process deficiencies).
Supplier	Prepare PFMEA with input from a multi-disciplinary team (Assembly, Manufacturing Engineering, Materials, Quality, Service, Suppliers, etc.) and identify KCC's for use in control plan.
Supplier	Adhere GM 1927 21 DFMEA PFMEA Gap Analysis Process and Transition Form (RPL Reduction summary) , including the categorization of risk, establishment of a top ten list, development of recommended actions, and update on a monthly basis to monitor progress. Present the evidences to meet GM 1927 37 PFMEA Audit
Supplier	Drive simple and inexpensive mistake / error proof devices into the process to help prevent and detect errors.
Supplier	Ensure that all failure modes and related severity numbers from DFMEA are listed in the corresponding PFMEA.
Supplier	Ensure that the current process controls and results of recommended actions on the PFMEA are listed on the control plan.
Supplier	Ensure that the PFMEA has a link with the manufacturing process flow chart and the control plan.
Supplier	Ensure that PFMEAs, Process Control Plans, and Standardized Work and Quality Checks sheets documentation are comprehensive, sufficient, and flow one from the other.
Supplier	If the process, material or manufacturing location changes, revise the PFMEA and re-evaluate the impact on severity, occurrence, and detection ratings.
Supplier	Ensure that subcontractors' PFMEAs are developed and RPL reduction activities are implemented.

Reference Documents:

- AIAG FMEA VDA Handbook or AIAG PFMEA 4th Edition and CG4338 GM 1927 03 SQ
- AIAG Advanced Product Quality Planning and Control Plan manual Process FMEA Checklist A-7
- GM 1927 21 DFMEA PFMEA Gap Analysis Process and Transition Form
- GM 1927 30 Quality Management Gap assessment
- GM 1927 36 Quality Management Elements Training (5 modules in SupplyPower)
- GM 1927 37 PFMEA Audit

Task Number: 13

Task Name: Process Control Plans

Task Owner: Supplier

Task Timing: Initial draft prior to Sourcing, updates parallel PFMEA changes and updates prior to PPAP

Task Description: Define the method to be used to control all KPC, PQC, AQC and DR's (at a minimum) through KCCs for parts being built for vehicle builds, Propulsion (GPS), and service applications to ensure customer requirements. Develop in line with the control plan format referenced in the AIAG APQP Manual.

Key Deliverables:

- Control Plan
- Individual Process Control Charts for all KPC, PQC, and DR
- GM 1927 16 Process Control Plan Audit

Customer for Deliverables: Supplier Quality, Manufacturing Engineer, Supplier

Necessary Inputs:

- Control plans on similar components
- Process Flow Chart
- PFMEA
- Special Characteristics
- Part-Specific Quality & Process CG (if applicable)
- Lessons Learned

Source of Input:

Supplier
Supplier
Supplier
Supplier/Engineer
SQE
SQE/Engineer/Supplier

Resources: Supplier and SQE

<u>Responsible:</u>	<u>Methodology:</u>
SQE	Verify that the supplier used the PFMEA and statistical data to determine what controls are necessary.
SQE	Verify the control plan is linked to the PFMEA, the Process Flow Chart and the Work Instructions. (Risk reduction updates done for PFMEA are linked to PCP and cascaded to Standardized Work. Documentation are comprehensive, sufficient, and flow one from the other)
SQE	Verify that the control plan covers each phase of the process, including re-work, receiving, in-process operations, packaging, labeling, material handling and shipping.
SQE	Ensure Part-Specific Quality & Process CG (if applicable) is applied to support Control Plan development.
SQE	Verify that the supplier updates the control plan as solutions to open issues are identified.
SQE	Verify that the supplier has individual process control charts for all KPC, PQC, and DR and that they have a means to store and recover this information for a period of 3 years.
SQE	Verify that pre-launch issues have been incorporated into the production control plan.
SQE	Walk the production floor and verify that the controls listed on the plan are in place and being used. Complete the GM 1927 16 Process Control Plan Audit as part of the audit of the production process.
SQE	Verify that Reaction plan(s) from the Process Control Plan are present, followed and effective.
Supplier	Develop a preliminary control plan using an existing control plan on a similar part. This first version of the control plan is then submitted with the quality portion of the supplier's bid package.
Supplier	Develop a pre-launch control plan for use on the first production parts shipped to assembly plants (GM 1927 28 Early Production Containment) and production control plan using the preliminary control plan as a foundation. The PFMEA and statistical data are used to determine which steps require additional control. The pre-launch control plan does not need to be separated from the production control plan. Pre-launch controls can be documented on the production control plan as long as they are clearly identified as such. Use the pre-launch control plan to validate the effectiveness of the production control plan. Apply AIAG checklist A-8
Supplier	Sample size and frequency should be determined based on risk and occurrence of the cause and ensures that the frequency adequately protects the customer by ensuring that the product represented by the inspection/test does not reach the customer before the results of these inspections/tests are known.

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Responsible:	Methodology:
Supplier	Develop individual process control charts for all KPCs, PQC's, and DR's based on the control plan using industry standard statistical process control techniques.
Supplier	Ensure the control plan is linked to the PFMEA, the Process Flow Chart and the Work Instructions. (Risk reduction updates done for PFMEA are linked to PCP and cascaded to Standardized Work. Documentation are comprehensive, sufficient, and flow one from the other)
Supplier	Identify and communicate any changes to the GM SQE.
Supplier	Update the control plan as solutions to open issues are implemented.
Supplier	Review control plans of subcontractors.
Supplier	Extend control plan to include installation at assembly plant (if supplier is responsible).
Supplier	Conduct layered audits to verify that the controls listed on the plan are in place and being used consistently.
Supplier	Validate, on a daily basis, that error proofing devices function properly.
Supplier	Develop effective Reaction plan(s) from the Process Control Plan.

Reference Documents:

- AIAG Advanced Product Quality Planning and Control Plan manual
- AIAG FMEA VDA Handbook or AIAG PFMEA 4th Edition and CG4338 GM 1927 03 SQ SOR
- AIAG Control Plan Checklist A-8
- IATF 16949 requirements
- GM 1927 16 Process Control Plan Audit
- GM 1927 28 Early Production Containment
- GM 1927 30 Quality Management Gap assessment (recommended tool)
- GM 1927 36 Quality Management Elements Training (5 modules in SupplyPower) (recommended)

Task Number: 14

Task Name: Early Production Containment Verification – EPC (GM 1927 28)

Task Owner: Supplier

Task Timing: Early Production Containment (GM 1927 28) start at PPAP non-saleable to meet APPV; containment until 90 days after SORP and meets exit criteria defined per GM 1927 28.

Task Description: Establish a containment plan during start-up and acceleration, so that any quality issues are quickly identified at the supplier's facility and not at the GM customer's facility. This procedure applies to all new and changed parts that require PPAP for start-up, acceleration or after shut down for extended period of downtime.

Key Deliverables:

- GM 1927 33 Early Production Containment Plan Audit
- Pre-Launch Control Plan
- GM 1927 39 Early Containment Exit Letter

Customer for Deliverables: GM Manufacturing or Assembly Facility, Supplier Quality, Purchasing

Necessary Inputs:

- Production Control Plan
- Early Production Containment (GM 1927 28) timetable

Source of Input:

Supplier / GM Customer

Resources: Supplier, SQE, Assembly Plant

<u>Responsible:</u>	<u>Methodology:</u>
SQE	GM 1927 28 Ensure Early Production Containment plan is available (agree on timing and exit criteria). Reinforce that failure to execute GM 1927 28, or shipment of a single defect during Early Production Containment GM 1927 28 period, may be considered for immediate CS2.
SQE	Verify that high RPLs, KCDS Designators are addressed by the pre-launch control plan.
SQE	Verify that the supplier used the PFMEA and statistical data to determine what additional controls are necessary (short term capability data on actual process or long term capability data on similar processes).
SQE	Verify any SPPS record or pilot issues are addressed by the containment process.
SQE	Review the supplier's containment process and pre-launch control plan using Early Production Containment audit GM 1927 33 at supplier plant as soon as Early Production Containment GM 1927 28 is implemented. Verification should include: • Actual drawings at Early Production Containment area with the latest change level. • Check frequency (100% or a documented lot sampling with GM approval). • Gauge operation instructions at Early Production Containment GM 1927 28. • Master parts or boundary samples to confirm inspection. • Inspection data for all components in Early Production Containment area. • Process in place to prevent shipment of material that has not gone through Early Production Containment GM 1927 28 verification. • Record of results.
SQE	Review supplier's initial Early Production Containment GM 1927 28 verification data to assess compliance to process intent.
Supplier	Develop an early production containment plan as specified in GM 1927 28 , including, but not limited to: • A separate inspection area whenever possible (may require checks within the process for components that are not available for check after assembly). • Process to ensure that all non-conformances are contained within the facility and prompt containment and irreversible corrective actions are applied if non-conformances are discovered. • Identification of the person responsible for the containment process. • 100% check of specific features for pre-pilot and pilot material, as required. • Use of green dots (signed by a designated senior management representative) on shipping labels to designate compliance. • Use of the Early Production Containment Plan until the agreed exit criteria is met.

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Responsible:	Methodology:
Supplier	Require compliance to GM 1927 28 from all subcontractors, as well as monitor and maintain their records. This will require that subcontractors pull ahead their Early Production Containment GM 1927 28 plan to allow the Tier 1 GM 1927 28 timing compliance.
Supplier	<u>Verification for Exit</u> Meet the defined exit criteria as per GM 1927 28. Supplier to request exit from Early Production Containment GM 1927 28 when eligible and provide supporting documentation and assessments on performance and corrective actions, if applicable, to the appropriate Customer representative. Documentation may be in form of I-chart (GM 1927 66) or other format as per GM 1927 28. Upon approval of EPC exit from GM SQ, upload signed GM 1927 39 Exit letter in SQMS Customer Specific Element Tab.
GM Assembly Center / SQ / plant QE	Verify that the supplier has met exit criteria per GM 1927 28, 4.0. (Minimum of two weeks supporting data.) Notify the supplier that they have met the criteria and that they are removed from Early Production Containment GM 1927 28 . Early Production Containment exit letter (GM 1927 39) optional.

Reference Documents:

- AIAG Advanced Product Quality Planning and Control Plan manual & Production Part Approval Process Manual
- CG4355 GM 1927 03 SQ Sub Tier Supplier Management
- GM 1927 11 APQP Kickoff
- GM 1927 17 Supplier Quality Processes and Measurements Procedure
- GM 1927 28 Early Production Containment Verification
- GM 1927 33 Early Production Containment Audit
- GM 1927 39 Early Production Containment exit letter
- GM 1927 66 CS1 and 2 I-chart

SQE	Drive adherence to PPAP submission date, as scheduled on GM 1927 02 APQP Timing Chart.
SQE	Review PPAP submissions as appropriate and confirm the PPAP status as per the AIAG PPAP manual and GM specifics.
SQE	Ensure adequate Action Plan is provided for PPAP without full approval.
SQE	Follow up action items on the Action Plan until full PPAP approval is achieved.
Supplier	Submit PPAP package according to AIAG PPAP requirements for the appropriate submission level on the date agreed on GM 1927 02 APQP timing chart.
Supplier	Verify compliance to the part-specific CG. Provide documentation of compliance with the PPAP submission (PPAP full approval requires compliance to CG4338 GM 1927 03 SQ SOR).
Supplier	Provide Action Plan in SQMS for SQE signature for any part submitted for less than approved status to achieve full approval. Provide any other appropriate signature dependent on issue as required (reference GM 1927 09 PPAP Action Plan Worksheet for required signatures). PDT should agree on Action Plans before submittal.
Supplier	EFFECTIVE January 1 st 2017, ALL “PPAP Action Plan” contents in SQMS are required to be in ENGLISH or if region-specific language is utilized, an English translation must be provided by the Supplier.
Supplier	Provide the samples applied to Matching verification with dimensional inspection results as per Global Part Submission Procedure for Matching GM10067 and GM 1927 32 Dimensional Report.
Supplier	The supplier shall document containment plans until process capability has been achieved.
Supplier	Shipment of parts for Saleable products is only allowed if the PPAP Status (SQMS PPAP Shipping Status) is Approved or Saleable with a valid expiration date. GM may choose to allow parts with a Non-Saleable Status to ship for a Saleable build but these parts need to be retrofitted before the product is delivered to the customer.
Supplier	Notify SQE before the PPAP re-submission if any process change occurs relative to the previous PPAP review.
Supplier	Guarantee the accomplishment of on time PPAP.
Product Eng. /PPO-SQ	Provides part quality sign-off through PPQP to support IVER/PVVB builds, regardless if tools are production tools or not.

Reference Documents:

- AIAG Production Part Approval Process manual and AIAG Statistical Process Control manual
- AIAG Measurement Systems Analysis manual and AIAG Advanced Product Quality Planning and Control Plan manual
- AIAG FMEA VDA Handbook or AIAG PFMEA 4th Edition and CG4338 GM 1927 03 SQ
- AIAG Quality System Requirements (IATF 16949 Standard)
- GM Worksheet GM1411 (reference for required signatures only)
- General Motors Commodity sign-off GM3660 (CG4816) (Available through Accuris (formerly IHS Markit) website)
- Generic Supplier ADV Process Tasks & Deliverables (GMW3600)
- Global Part Submission Procedure for Matching GM10067
- GM Customer Specific Requirements – IATF 16949
- Part-Specific Quality & Process CG (if applicable)
- GM 1927 02 APQP Timing Chart
- GM 1927 09 PPAP Action Plan Worksheet
- GM 1927 32 Dimensional Report

DEFINITIONS:

Approved – Meets all requirements as defined in AIAG PPAP, Latest edition

Non-Saleable - Does not meet all the requirements of AIAG PPAP, Latest edition, but has been deemed acceptable for use on **Non-saleable** vehicles. This status requires approval from the the applicable customer roles as defined in 1927 09 for all PPAP elements not fully approved.

Pending Evaluations – Indicates that the supplier has not submitted for any status or that the customer has not accepted the status that has been submitted by the supplier.

Rejected – Does not meet the requirements as defined in AIAG PPAP, latest edition and is not deemed acceptable to be used in the customer facilities.

Saleable - Does not meet all the requirements of AIAG PPAP, Latest edition, but has been deemed acceptable for use on **Saleable vehicles**. This status requires approval from the the applicable customer management roles as defined in 1927 09 for all PPAP elements not fully approved.

Submission: the act of presenting PPAP documentation for the status decision either directly into SQMS, through e-mail, or visually (in person or in virtual meetings). Submission technique must be agreed to by SQE. Copies of all PPAP documentation, regardless of submission technique must kept on file at the supplier.

Task Number: 16

Task Name: Run at Rate (R@R) / Seat Complete (SC) Build at Rate (B@R)

Task Owner: Supplier

Task Timing: CRV Assessments and at Run at Rate / Build at Rate

Task Description: Verify that the supplier's actual manufacturing process, while operating under normal operating conditions and under total customer requirement, is capable of producing components that simultaneously meet Quality and Daily Contracted Capacity

Key Deliverables:

- Input of all required data in SQMS Run at Rate module or Completed B@R Workbook for each Seat Complete Line, and Sub-assemblies as needed.
- Verification Supplier can meet Supplier Daily Capacity (SDC) in one day

Customer for Deliverables: Supplier Quality and Purchasing

Necessary Inputs:

- PPAP documentation
- GM's daily requirement (LCR)
- Supplier's Contracted Tool Capacity (SDC)
- Manufacturing Block Diagram
- Production & Quality Performance Data
- Sub-tier Suppliers written confirmation of quality, capacity and delivery for all sub-components

Source of Input:

Supplier
Buyer
Supplier & Buyer (contract)
Supplier
Supplier
Supplier

Resources: SQE, Buyer, DRE, and Supplier

Responsible:	Methodology: (refer to R@R and B@R diagrams at end of the task)
GM Buyer /SQE	Conduct the APQP Kick-off meeting and confirm with GM Purchasing Buyer that the Daily contracted capacity is accurate and is greater than or equal to GM's daily requirement (LCR). Confirm part grouping is correct. Confirm with GM Buyer if staged Capacity is allowed in the contract. Utilize escalation process as required, to support resolution of any outstanding contract issues. For SC Build at Rate: Initial and updates required 2 weeks prior to each CRV event - Review preliminary B@R workbook: SC Build at Rate Analysis top sections, contracted SDC, and Section A, GM 1927 35d B@R Form Sections 1&2, Subcontractor plan for every part (PFEP) Form, Shared Capacity, and Mfg. Block Diagram - Update CRV Status (Red/Green) to communicate "Risk"
GM Buyer/SQE	At the APQP Kick-off meeting evaluate the supplier's initial Manufacturing plan including Plant Layout and Manufacturing Block Diagram (including all processes, tools, banks, buffers, manpower, planning gross and net rates, first time quality assumptions and change over times, if applicable.)
GM Buyer/SQE/DRE	Conduct the CRV Assessments and evaluate the Supplier's latest Manufacturing plans including Plant Layouts, the Manufacturing Block Diagram, SQMS Capital, Tooling, Detail Process, Sub-Process inputs (including shared capacity). At each CRV, Publish results in SQMS should reflect latest performance data, GM 1927 35d (for Seat Complete B@R) Work Book including Shared Capacity sheets when applicable.
GM Buyer/SQE/DRE	Follow escalation process for any capacity issues discovered during the CRV Assessments as required.
SQE	Determine and communicate customer or supplier monitored Run @ Rate requirements to Tier 1.
SQE	Review Tier 1 Suppliers Sub-Tier Risk Assessment (GM 1927 07a Risk Assessment or equivalent) to confirm the identification of "Critical" Sub-Tier suppliers.
SQE	Verify that Tier 1 suppliers have had all of their sub-tier suppliers confirm in writing and provide documented evidence (GM 1927 35a, GM 1927 35d or equivalent) to demonstrate their ability to meet quality and delivery requirements (including service parts).
SQE	Attend all Customer monitored Run @ Rates or Seat Complete Build @ Rates. Ensure all Run @

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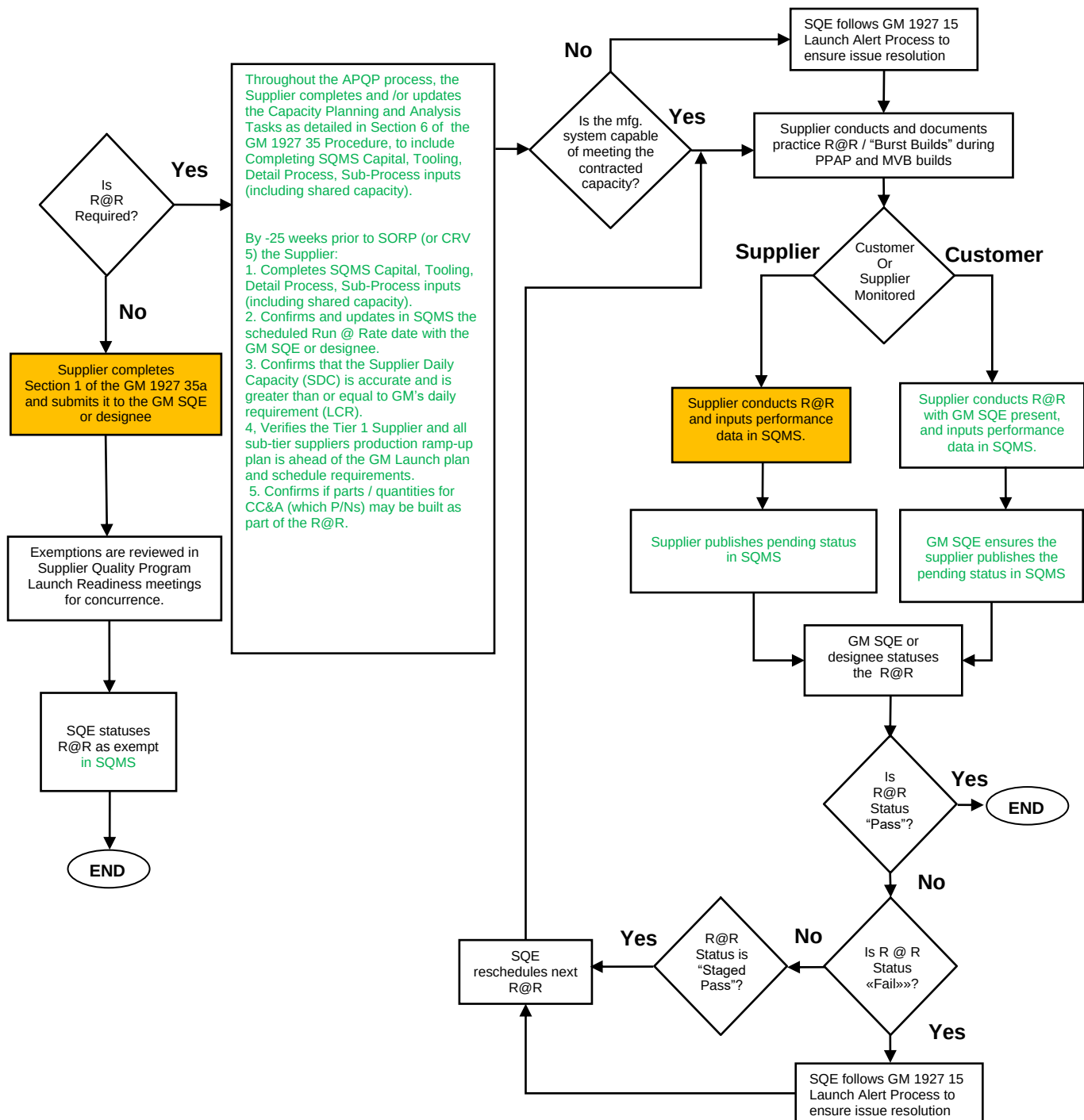
	Rates or Build @ Rates are performed under normal production operating conditions including planned production, maintenance, and support staffing levels and production rates.
SQE	Review all Run @ Rate or SC Build @ Rate/ SQMS System documentation for accuracy and completeness. Evaluate the capacity of all potential constraint (all operations that are 90% utilized or above) and final assembly operations, in addition to shared capacity operations. Status the Run @ Rate or SC Build @ Rate (Customer or Supplier Monitored) based on the installed capacity meeting the contract requirements per the GM 1927 35 Run @ Rate Procedure or per GM 1927 35c SC Build at Rate Procedure, whichever is applicable.
SQE	Follow the correction action and escalation processes for all “FAIL” Run @ Rates or SC Build @ Rates (Follow the requirements of the GM 1927 35c Build @ Rate Procedure regarding timing)
Supplier	At the APQP Kick-off Meeting, confirm with the SQE and the GM Buyer that the Supplier Daily Capacity is accurate. Supplier to confirm in SQMS R@R module. Supplier to create Master Process and identify and escalate any contractual concerns: - SDC is accurate and greater than or equal to GM’s daily requirement (LCR). - Less Finish / Make From are contracted correctly. - Confirm part grouping and group contract are correct. - Confirm if staged capacity is correct. For SC Build @ Rate: Initial and updates required 2 weeks prior to each CRV event; Review preliminary SC B@R workbook: Build at Rate Analysis top sections, contracted SDC, and Section A, GM 1927 35d B@R Form Sections 1&2, GM 1927 25 Subcontractor status matrix, Shared Capacity, and Mfg. Block Diagram; Update CRV Status (Red/Green) to communicate “Risk”
Supplier	Provide the initial Manufacturing plan including Plant Layout and Manufacturing Block Diagram (including all processes, tools, banks, buffers, manpower, planning gross and net rates, first time quality assumptions and change over times, if applicable.) at the APQP Kick-off meeting.
Supplier	Provide/update the latest Manufacturing plans including Plant Layouts, the Manufacturing Block Diagram, SQMS Capital, Tooling, Detail Process, Sub-Process inputs (including shared capacity). At each CRV, Publish results in SQMS should reflect latest performance data. For Seat Complete use GM 1927 35d Worksheet including Shared Capacity sheets when applicable, at each CRV Assessment. The capacity analysis must show supplier’s capability to meet SDC.
Supplier	Complete Sub-Tier Supplier Risk Assessment (GM 1927 07a Risk Assessment or equivalent). Communicate customer or supplier monitored Run @ Rate requirements to Tiered Suppliers.
Supplier	Communicate SDC, LCR and MCR volumes to Sub-Tier Suppliers.
Supplier	Ensure understanding of the R@R or SC B@R/ SQMS System “PASS” performance requirements.
Supplier	Conduct preliminary R@Rs and/or burst builds /training build events prior to the planned R@R or Seat Complete (SC) B@R to confirm readiness. Collect and analyze production and quality data including hourly production counts, scrap and/or first time quality (FTQ) data, scheduled and unscheduled downtimes, operational cycle times, and changeover times. The PPAP run of parts should be planned to one of the trial runs.
Supplier	Schedule and conduct actual Run @ Rate or SC Build @ Rate, Burst and Training events with GM SQE.
Supplier	Conduct all Run @ Rates or SC Build @ Rates under normal production operating conditions including planned production, maintenance, and support staffing levels and production rates. Document all required production and quality data for all potential constraint and final assembly operations, as well as shared capacity operations.
Supplier	Complete all appropriate sections of the GM 1927 35a or GM 1927 35d Workbook. Submit the workbook and review the results of the Run @ Rate or SC Build @ Rate with the GM SQE.
Supplier	Complete capacity verification of all Sub-Tier Suppliers and submit all appropriate written documentation (Subcontractor PFEP) to the GM SQE. For SC Build @ Rate once approved the documents must be loaded into SQMS under the "Customer Specific Requirements" element.
Supplier	Complete and follow the corrective action plan with timing, to achieve the Contracted Daily Capacity (SDC) for all “FAIL” Run @ Rates or SC Build @ Rates. Submit and review this plan with the GM SQE within two days (48 hours) of the completion of the failed Run at Rate or SC Build @Rate.

Reference Documents:

- GM 1927 07a APQP Risk Assessment subtier
- GM 1927 25 Subcontractor status matrix
- GM 1927 35 Run at Rate Procedure
- GM 1927 35a Run at Rate Workbook
- GM 1927 35b Capital and Tooling Matrix for Reference Purposes

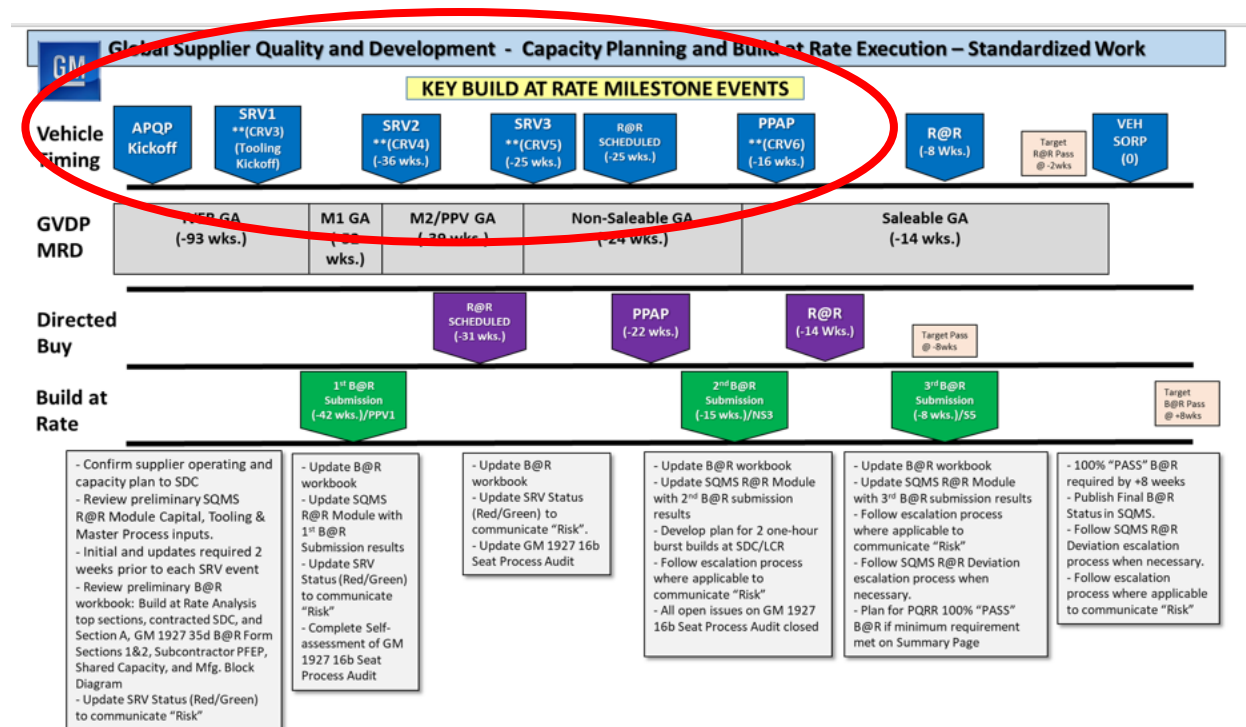
- GM 1927 35c Build at Rate Procedure - when applicable
- GM 1927 35d Build at Rate Workbook - when applicable

R@R Methodology



B@R Methodology

This process is for use by all Seat Complete Suppliers.



Task Number: 17

Task Name: Lessons Learned

Task Owner: Supplier/ GCL (GRA)

Task Timing: Commodity Key Stakeholders Meeting through Launch

Task Description: Maximize the knowledge gained from previous programs and not revisit the same quality or design issues. Lessons Learned includes a standardized closed loop learning process (Global Read Across) for closing out Safety specific issues (Safety SPPS, Safety PRTS, PRD, and Pre-emptive events).

Key Deliverables:

- Defined Lessons Learned process
- Evidence of learnings shared within the facility
- Evidence of learnings shared across the organization, includes all functions (Engineering, Manufacturing, Quality, etc.) & facilities globally
- Evidence of learnings shared with GM for continuous improvement
- Evidence of a completed lessons learned document (when applicable) as a tool to communicate lessons learned
- Evidence of DFMEA, PFMEA, and baseline design guidelines or process best practices updated with Lessons Learned
- Compliance to Part-Specific Quality & Process CG for key commodities
- Evidence of Part-Specific Quality & Process CG updated with Lessons Learned throughout the APQP process
- **Customer for Deliverables:** SQ, Engineering, PDT (Product Development Team), Supplier
- Completed and Distributed GM 1927 77 Global Read Across Form and Procedure GRA
- Lessons Learned Database of GRA items stored in Auros System

Customer for Deliverables: SQ, Engineering, PDT (Product Development Team), Supplier

Necessary Inputs:

- Existing Lessons Learned for same commodity
- Engineering Knowledge for commodities
- Process knowledge for SMT / Commodity
- GM 1927 77 GRA Form and Procedure
- Live Data GRA Report / GRA Tracker

Source of Input:

SQE
Engineer
SQE / GCL / GPL / Supplier
SQE / GCL / GPL / Supplier
SMT / Commodity GCL

Resources: Supplier, SQE, GCL, GPL, GPM, Engineering, Auros Lessons Learned Database

Responsible:	Methodology:
SQE	Lessons Learned Update : (Commodity Key Stakeholders Meeting): • Ensure Part-Specific Quality & Process CG (if applicable) is included in the RFQ package. • The direction for the supplier is to use the information during design and process development. (Technical Review): • Provide an overview to suppliers on the lessons learned process and how it fits into APQP. • Ensure that the potential suppliers understand the Part-Specific Quality & Process CG (if applicable). (Kick-off Meeting): • Review any additional information that has been obtained on lessons learned from local or regional databases. • Ensure Lessons Learned issues are incorporated into the APQP Open Issues List GM 1927 05. • Ensure that the suppliers understand the Part-Specific Quality & Process CG (if applicable). • The goal is a zero tolerance for defects.
SQE	Ensure in each of the subsequent Valve Assessments that the supplier executes Lessons Learned process (Global APQP task #17) and apply to the Part-Specific Quality & Process CG (if applicable).
SQE	Global Read Across: • Issue / Provide Supplier Responses for all Global Read Across activities • Review Lessons Learned Auros Database for key Lessons Learned • Support GCL completion / update of the GRA Form as necessary

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GPM / (Global Process Manager) SMT Manager	Global Read Across: - Review and Distribute Level B&C GRA Documents to SMT's - Status GRA Tracker for Level B&C GRA Activities - Perform Follow-up with SMT to ensure completion of all Level B&C GRA's - Perform Follow-up with SMT on GRA data entered into Auros (Lessons Learned Database)
GCL (Global Commodity Lead)	Global Read Across: - Work with SQE / Supplier to complete the GRA Form for approvals and entry in Auros - Hold and Support GRA Peer Reviews and communicate SMT specific GRA Lessons Learned - Ensure proper completion of GRA form GM 1927 77 Global Read Across Form and Procedure - Upload GRA data into Auros Database - Update SOR / CG Documentation as determined by SQ Leadership and Peer Review
Supplier	Lessons Learned Update : Review solutions to issues identified in the CRV 0 and new items added to the list: At each CRV (Run @ Rate): during the MVBs As solutions are identified, DFMEA, PFMEA, Flow Chart, and Control Plan are reviewed and updated. Standardized Work should be updated as well when needed.
Supplier	Adhere to the Part-Specific Quality & Process CG, (if applicable).
Supplier	Ensure a formal process is in place to update FMEA and control plans based on Lessons Learned.
Supplier / SQE / DRE	Review Lessons Learned from the project and incorporate identified key items into Part-Specific Quality & Process CG (if applicable) to ensure that future programs do not have the same issues.

Reference Documents:

CGXXXX GM 1927 03a Part-Specific Quality & Process

GM 1927 77 Global Read Across Procedure and Form GRA

Global Read Across Training (LMS Course)

AIAG FMEA VDA Handbook or AIAG PFMEA 4th Edition and CG4338 GM 1927 03 SQ SOR

Section 2

Global Current

Document Usage Guidelines by Global Region

F = Reference Only (Information only, Nothing to fill out)

NA = Not applicable for specified region

MD = Mandatory Document (document type required, specific format required)

MI = Mandatory Information (document type required, format may vary, minimum content mandatory)

RD = Recommended Document (document recommended, specific format required)

RI = Recommended Information (document recommended, format may vary)

Document	Number	GMIO	GMNA	GM South America	GMM
Process Control Plan Audit (PCPA)	GM 1927 16	MD	MD	MD	MD
RPL Reduction Summary (DFMEA/PFMEA GAP analysis Process and Transition Form)	GM 1927 21	MD	MD	MD	MD
Quality Management Gap assessment	GM 1927 30	RD	RD	RD	RD
Quality Management Key Elements Training (5 Modules)	GM 1927 36	RD	RD	RD	RD
Controlled Shipping Level 2 Assessment Matrix	GM 1927 51	MI	MI	MI	MI
Controlled Shipping Level 1 Assessment	GM 1927 52	M	M	M	M
Controlled Shipping CS1/CS2 Verification Exit Checklist	GM 1927 54	RD	RD	RD	RD
Controlled Shipping Level 2 (CS2) Exit Survey (Supplier)	GM 1927 58	MI	MI	MI	MI
Controlled Shipping Kick-off Meeting Worksheet	GM 1927 59	MD	MD	MD	MD
Global Emerging Issue Alert	GM 1927 65	MI	MI	MI	MI
I-Chart Controlled Shipping	GM 1927 66	RI	RI	RI	RI
New Business Hold Process Flow	GM 1927 67	F	F	F	F
Drill Deep & Wide SPPS Read Across Matrix	GM 1927 69	MI	MI	MI	MI
Quad Report	GM 1927 71	MD	MD	MD	MD
Shutdown/Start-up audit	GM 1927 73	MD	MD	MD	MD
Disruption Agenda Format	GM 1927 75	MI	MI	MI	MI
Global Read Across Form	GM 1927 77	MD	MD	MD	MD
Step Down Chart	GM 1927 78	MD	MD	MD	MD
Supplier Corrective Action Plans	GM 1927 79	MI	MI	MI	MI
5 Whys 3 Times (Drill Deep & Wide Worksheet)	GM 1927 84	MI	MI	MI	MI
Drill Deep & Wide Audit	GM 1927 85	MI	MI	MI	MI
Drill Deep & Wide Root Cause Workbook	GM 1927 86	MI	MI	MI	MI

All documents can be found at: SupplyPower library

Task Number: 1

Task Name: Continuous Improvement

Task Owner: Supplier

Task Timing: Throughout the life cycle of the product

Task Description: Defines the supplier's responsibility to have an on-going process for continuous improvement of the product and related manufacturing processes; The intent is to reduce variation and to ensure process stability and capability over time. There must be a performance attitude for zero defects.

Criteria for Application: Applies to all suppliers and products for the life of the part.

Key Deliverables:

- First Time Quality Trend Chart (I-Chart recommended GM 1927 66)
- Problem solving capability to establish root cause
- Process Capability Continuously Compliance to Supplier Quality SOR (CG4338 GM 1927 03 and or CGxxxx GM 1927 03a name)
- GM 1927 84 Drill Deep Worksheet
- GM 1927 21 Gap analysis process and transition form (RPL Reduction Summary)
- PFMEA, Process Control Plan, Operator Standardized Work Instruction – Revised
- GM 1927 79 Supplier Action Plan Form
- GM 1927 69 Drill Deep Wide Matrix

Customer for Deliverables:

GM Assembly/Manufacturing Plants
Supplier Quality Engineer
Supplier Manufacturing Management

Necessary Inputs:

- Internal (FTQ, Process Capability, etc.)
- External Data Analysis (9-Panel, Warranty data, etc.)
- Parts Specific CG (CGxxxx GM 1927 03a name)
- Commodity Audits (GM 1927 16a name)
- Lessons Learned

Source of Input:

Supplier
SupplyPower
SQE / SupplyPower
Supplier

Resources: Supplier, SQE, Assembly Plant

<u>Responsible</u>	<u>Methodology:</u>
SQE	Monitor and audit the supplier's compliance to Continuous Improvement requirements, and also continuous compliance to Part specific CG / PCP Audits
SQE	Communicate specific concerns to the supplier for non-conformance, review action plans (as required), escalate concerns to leadership (if required), and require the supplier to implement countermeasures (if necessary).
Supplier	Identify Special Product and Process Characteristic • Supplier shall identify product and process characteristics that require KCDS approach • Inputs for this identification: a. Customer Drawing Information b. PFMEA c. Process/Product Capability d. Past or potential issue (9- Panel, Warranty Report, Internal Report)
Supplier	Establish Process Controls for items identified • Include KCDS items in Process Control Plan (PCP). • Establish Key Control Characteristics (KCC) List. • Manage process capability and performance 1. Variation control (SPC) 2. Corrective action & process improvement

	Update process capability / performance studies (FTQ, Cp/Cpk)
Supplier	Measure First Time Quality - First time quality (FTQ) is a measure of a process' ability to make quality parts without scrap or rework. The rework of parts can result in undesirable variation and may take parts out of process if the rework was not initially comprehended in the process. - FTQ should be measured at: - End of line FTQ measurement (at a minimum); - Quality gates or verification stations for key processes - Quality gates or verification stations should include: Alarm limits established (e.g. 1 defect per hour) to drive immediate containment upstream in the process; A Pareto of failure modes for a specific timeframe; FTQ trend chart (I-Chart format recommended – GM 1927 66)
Supplier	Continuously demonstrate process capability compliance to requirement in CG4338 GM 1927 03 Supplier Quality SOR (and or CGxxxx GM 1927 03a name), and have an ongoing process to reduce variation on characteristics identified as KPC's
Supplier	Effective Root Cause Analysis (Compliance with SPPS) and Corrective Action Verification (Compliance with GM 1927 17 Supplier Quality Processes and Measurements Procedure) - The supplier is required to perform an effective analysis of technical root cause. - Systemic root cause to be analyzed through the Drill Deep process - use worksheet GM 1927 84 - Verification and validation of corrective action should demonstrate the ability to turn the problem on and off.
Supplier	RPL Reduction Process (to have a Standardized, formal and documented RPL reduction process which includes: - A cross- functional team Define a Standardized Reverse PFMEA audit concept then meets regularly and: - Confirm the current failure modes have the identified methods and controls in place. - Experiment with the station in order to try to find new failure modes. - Document audit findings in an action plan with champion and dates to complete. - Findings are driven back into Control Plan, PFMEA, Process Flow, Work Instructions as applicable.
Supplier	Update and revise Quality Documentation – PFMEA, Control Plan, and Standardized Work. The revisions are not only required when there are design or process changes, but are based on internal or external data. - Internal data should drive quarterly updates at a minimum - External data should drive monthly updates at a minimum
Supplier	Supplier should have a process in place, which would drive lessons learned into current and future application of FMEAs and Control Plans. The process should apply lessons learned by a look across of similar processes within the facility and within the company (Recommend Drill Wide Matrix – GM 1927 69) and compliance to Part Specific CG / PCPA shared by SQE)

Reference Documents:

- AIAG SPC Documents
- CG4338 GM 1927 03 Supplier Quality SOR and/or CGxxxx GM 1927 03a name Commodity /Part Specific
- GM 1927 16 Process Control Plan Audit (PCPA)
- GM 1927 16a name Commodity Audit
- GM 1927 17 Supplier Quality Processes and Measurements Procedure
- GM 1927 21 DFMEA PFMEA Gap Analysis Process and Transition Form
- GM 1927 30 Quality Management Gap assessment (recommended tool)
- GM 1927 36 Quality Management Elements Training (5 modules) (recommended)
- SPPS Records
- GM 1927 66 I-chart
- GM 1927 69 Drill Deep Wide Matrix
- GM 1927 79 Supplier Action Plan Form
- GM 1927 84 Drill Deep Worksheet
- FMEA,
- KCDS (GM1805, GMW15049)

- Task Number:** 2
- Task Name:** GM Quality Performance Requirements (QPR)
- Task Owner:** Supplier
- Task Timing:** Throughout the life of the product
- Task Description:** Suppliers are required to obtain and maintain compliance to GM QPR which include the following criteria:
- 1) IATF 16949 Certification
 - 2) Number of Field Actions
 - 3) Number of Major Disruptions
 - 4) Number of Quality SPPS in 6/12 months
 - 5) Severity Score for 6/12 months

Key Deliverables:

- If the Supplier is not IATF 16949 Certified - Action plan for IATF 16949 certification.
- If the supplier is IATF 16949 Certified and not meeting GM QPR they must submit to GM SQE an Action Plan to address the deficiencies. The Action Plan must be approved by GM SQE.

Customer for Deliverables: Supplier Quality

Necessary Inputs:

- IATF 14969 Certification
- Sourceability / Impact detail report

Source of Input

- IATF data base
- SCMS

Resources: SQE and Supplier

<u>Responsible</u>	<u>Methodology:</u>
Supplier	The supplier needs to demonstrate that they have a tool to assess their quality system and compliance to Quality Performance Requirements. The supplier's quality system assessment should be to further develop sustainable quality systems and for Continuous Improvement.
Supplier	If the supplier is not meeting GM QPR, then supplier must create an Action Plan to meet the requirements. The Action Plans must be complete with appropriate detail, timing, ownership, etc. The Actions Plan shall be approved by GM SQE.
SQE	Monitor the Supplier Action Plan on a timely basis.
SQE	Failure to implement all Action Plan's items, could result in the SQE issuing a <i>Program Management SPPS</i> .
Supplier	Implement and verify effectiveness of all the action plan on a timely basis.
Supplier	Keep the customer updated regarding implementation status and any issues or concerns with implementation.

Reference Documents / Tools:

- GM 1927 36 Quality Management Training (5 modules in SupplyPower) (recommended tools)
- GM 1927 30 Quality Management Gap assessment (recommended tools)
- IATF 16949 Requirements
- GM 1927 78 Step Down Chart

Task Number: 3

Task Name: Shutdown / Startup Activity

Task Owner: GPSC – Current Supplier Quality

Task Timing: Throughout the life cycle of the product

Task Description: The purpose of the Shutdown/Startup Activity (SD/SU) is to implement standardized work for both the Supplier Quality Group and its suppliers (Production, Consignment / Bailment), specifically focused around extended shutdown periods >5 calendar days and subsequent startups. Examples include: Corporate (“summer”) shutdowns, Christmas shutdown, etc., but the task and tools can be used by any supplier when they are faced with any extended shutdown/startup > 5 calendar days, or other events such as the elimination and/or restarting of a production shift or an extended weekend.

The goal of the Shutdown/Startup Activity is ZERO major disruptions caused by a supplier’s shutdown and subsequent restart of operations.

Key Deliverables:

- GM 1927 73 Shutdown audit form completed
- GM 1927 79 Supplier Corrective Action Plan for High risk suppliers

Criteria for Application:

- At the discretion of the SQE, extended shutdown periods, such as the “Summer Shutdown”, “Christmas Shutdown”, Extraordinary Production Shutdown >5 consecutive days.
- Selected high-risk suppliers based on established criteria. (Examples and not limited to are the potential high risk suppliers criteria could include: Critical/Special Process (Heat Treating, Plating, Coating, etc.), Emerging Issues, Plant Disruptions, High or High Cost Warranty, Tool Moves, Process Change, Launch Issues, CS1 or CS2 Supplier’s, Program Management SPPS’s, High Financial Risk suppliers rated Red, new supplier to GM, and others)

Customer for Deliverables:
SQE
SQ Leadership
Buyer

Necessary Inputs:

Shutdown/Startup Audit Form (GM 1927 73) + Instructions
Master Dot Action Plan (GM 1927 79)
High Risk Supplier Listing with CEO & Plant Manager,
Names, e-mail addresses (reasonable size)

Source of Input:

Supply Power
Supply Power
SCMS

Resources: SQ Process/Data Group (where applicable)

Task Name: Shutdown / Startup Activity (Continued)

<u>Responsible</u>	<u>Methodology:</u>
SQ Coordinator/SQ Leader	• Holds an initial conference call with the representatives to review objective timing, the process and key dates. • Identifies key criteria for selecting targeted suppliers for the SD/SU Audit activity, as suggested by customer plant ▪ Previous evidence or concern of poor startup at the supplier • Documents the SQE’s standardized work as part of the SD/SU Audit activity.
SQ Team	• Identifies the suppliers to be targeted for the SD/SU activity based on the criteria provided • Assigns SQE’s to perform the standardized SD/SU work.
SQE	• Obtains the contact information for the assigned supplier(s), including the CEO and Plant Manager name, phone and e-mail address. • Sends the standardized communication/e-mail to the supplier. THIS SHOULD BE OPTIONAL (SEND THE AUDIT TO THE SUPPLIER SINCE SOME SQE TEAMS CONDUCT ON SITE REVIEW ONLY)

Supplier	• Performs a self-assessment of their shutdown/startup readiness using the SD/SU audit and supporting documentation provided by GM. • Completes the SD/SU audit form and returns to the assigned SQE and GSC Command Center for Consignment / Bailment suppliers with appropriate supporting documentation/evidence, or completes the web-based SD/SU online survey (if available). • Develops an action plan to resolve any issues identified in the self-assessment and implements containment and/or corrective actions. Sends the action plan to the assigned SQE and GSC Command Center for Consignment / Bailment suppliers.
SQE	• Judge the supplier's performance (R-Y-G) self-assessment on the shutdown/startup audit form or survey, looks for evidence to support the supplier's self-assessment, and challenges the supplier if no evidence is provided. • Gives the supplier an overall R-Y-or G rating and updates the supplier's status on a master tracking list • For Yellow suppliers, at a minimum, the SQE contacts the supplier to obtain clarification/evidence or further feedback on the open issues, and to request a supplier action plan if required. • For Red Suppliers, the SQE does an onsite review at the supplier location to resolve any red issues and review the supplier's action plans. • Places the supplier into Early production containment (GM 1927 28) verification or CS2 until all open issues are resolved. • Reviews the supplier's progress in resolving open issues and in implementing the action plan • Updates the supplier's R-Y-G overall status on the master-tracking list.
Supplier	• Reviews and rolls out GM Shutdown/Startup presentation internally. • Implement GM 1927 28 Early production containment or CS2 as required.

Additional Information:

- Previous Shutdown Audit, if applicable.
- Shutdown plan presentation.

Reference Documents:

GM 1927 28 Early Production Containment
GM 1927 73 Shutdown/Startup Audit Form
GM 1927 79 Master Dot Action Plan

Task Number: 4

Task Name: Process Control Plan Audit & Part / Process Specific Audits

Task Owner: Supplier Quality Engineer

Task Timing: Throughout the life of the product

Task Description: The purpose of the Process Control Plan Audit is to ensure the supplier is following the documented Process Flow Diagram and executing the Process Control Plan that was the basis for PPAP. It is also to ensure the supplier is maintaining and updating said documents, including Process Flow Diagrams, PFMEAs, and Process Control Plans, while executing continuous improvement.

The purpose of the Part/ Process Specific Audits is to ensure that the supplier has specific controls in the process that are not covered in the Process Control Plan Audit. The Part / Process Specific Audits are designed to be used in combination with the Process Control Plan Audit.

Key Deliverables:

- GM 1927 16 Process Control Plan Audit
- Part Specific or Process Specific Audits (GM 1927 16a *name*)
- GM 1927 79 Supplier Action Plan, if required, approved by SQE

Customer for Deliverables: SQE, SQ Manager, Supplier

Necessary Inputs:

SPPS Analysis Summary
Supplier's Internal Data Analysis
Process Flow Diagram
PFMEA
Process Control Plan

Source of Input:

SPPS records
Supplier
Supplier
Supplier
Supplier

Resources: Supplier Representative, SQE

<u>Responsible</u>	<u>Methodology :</u>
SQE	Decides or is assigned to visit a supplier to: Review a specific issue/concern Review an overall performance concern Gain a better understanding of the supplier's processes, etc.
SQE	Reviews appropriate external performance data. Data should include 9 Panel Charts, SPPSs, or a specific part/process concern.
SQE	Contacts supplier's Quality Manager to explain the nature of the visit and identify specific expectations relative to documents (such as part print, FMEA's, Process Control Plan, etc.) that need to be ready for review.
SQE	Identifies which product or process requires an audit. Chooses appropriate audit or audits to conduct.
SQE	Ensures that appropriate supplier representatives participate in the audit. Ensures the Supplier understands documented concerns/non-conformances. Implements appropriate action, if necessary, to ensure conformance to requirements and reduce risk of major non-conformances; such as containment, CS1 or CS2. Summarizes expectations relative to: Immediate corrective actions Action plans, format and due dates Follow-up time frame Additional requirements and expectations
Supplier	Supplier implements any additional internal process controls required to reduce the risk of major non-conformance immediately, to insulate the GM Manufacturing facility or the Containment Gate.

Supplier	Provides requested documents and participates in Audit. Understands concerns/non-conformances identified by the SQE. Implements any immediate containment/addition process controls as directed. Develops, documents, and provides action plan with appropriate detail, timing, ownership, etc. Implements action plan on a timely basis.
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Reference Documents:

GM 1927 16 Process Control Plan Audit
 GM 1927 79 Supplier Corrective Action Plan

Task Number: 5

Task Name: SQ Creativity Team Support

Task Owner: GPSC – Supplier Quality

Task Timing: On-going process specific to Creativity Team

Task Description: SQ Creativity Team (CT) support is a process to drive quality in the Creativity Teams by providing the voice of Supplier Quality into the GPSC sourcing process. Through the creation of CT quality action plans, Global Commodity Lead's (GCL) engage the CT in strategic and up-front activities to drive performance improvement on the Sourceability Report (CTBL). The SQE / Global Commodity Lead (GCL) assigned to the Creativity Team provides the voice of the customer through feedback from Supplier Quality into the Creativity Team.

Key Deliverables:

- Updated Sourceability Report
- Supplier Performance Improvement
- CT Metrics Tool
- Step-down Charts (GM 1927 78) Optional

Customer for Deliverables: Supplier, Supplier Quality, Creativity Team.

Necessary Inputs:

CT Performance Report (9 Panel)
Creativity Team Sourceability Report
SPPS detail review
Supplier's Internal Data Analysis
CT Roll out schedule
IATF 16949 Certification
Quad Report (GM 1927 71)
PCPA (Process Control Plan Audit – GM 1927 16)
PSA (Potential Supplier Assessment – GM 1927 08)

Source of Input:

Supply Power
SCMS
SCMS
Supplier
CT
Supplier/Certification Body
Supply Power
Supply Power
Supply Power

Resources: Global Commodity Lead (GCL)/ SQE – Global Champion & Regional Assigned CT Support, Global / Regional Buyers, CT members, Global / Regional Commodity Manager, Global / Regional Commodity Director, Supplier

Responsible	Methodology:
SQE/GCL	SQ CT Global Champion reviews CT necessary inputs.
SQE/GCL	SQ CT Global Champion involvement within the CT to determine Performance Improvement and the commodity key drivers.
SQE/GCL	SQ CT Global Champion participates in the CT meetings, understands the CT business plan, presents specific CT supplier quality and report out in Supplier's quality status.
SQE/GCL	SQ CT Global Champion coordinates required new supplier audits with their SQ counter-parts in the other regions and own – supports the GPSC sourcing process (RFQ-Target Setting-Final Recommendations)
SQE/GCL	SQ CT Global Champion document the necessary sourcing quality documents in SCMS, as specific to the CT that is assigned (PSA, PCPA and GM 1927 71 Quad Report)
SQE/GCL	SQ CT Global Champion reviews sourcing packages within the CT (RFQ-Target Setting-Final Recommendations).
SQE/GCL	SQ CT Global Champion uses the CT Meetings for poor performing supplier review
SQE	The DUNs' owning SQE uses Quality Management tools to support supplier to improve quality performance.
Commodity Manager (purchasing)	Provides CT agenda for each meeting, reviews CT business plans, rollout schedule (monthly), and CT performance.
Commodity Manager (purchasing)	Coordinates the sourcing package review before the GPSC Sourcing meeting within the CT members (RFQ-Target Setting-Final Recommendations) which supports Performance Improvement Provides SQ agenda time to present CT quality data and supplier reviews.
SQE/GCL	SQ CT Global Champion attains required sourcing approvals or rejection for a sourcing decision

SQE/GCL	SQ CT Global Champion works with the buyer to ensure red suppliers who will NOT be green by the time of the sourcing recommendation are NOT included in the Pre-RFQ.
SQE/GCL	SQ CT Global Champion must communicate and coordinate the proper documentation (regional SQ support) for Red Quality Sourcing Decision's – Quad Report (Quality Business Case) must be approved by the Executive SQ Director located in the supplier region before the GPSC ISC Sourcing meeting.
Supplier	Poor performing suppliers with the agreement of the CT will present at the CT Meetings their specific corrective actions to return to quality "Green" status on the Sourceability Report. Supplier's leadership provides documents and drives the improvement process.
SQE/GCL	SQ CT Global Champion / regional SQ support - reviews supplier's quality performance
SQE/GCL	SQ CT Global Champion / regional SQ support - provides the Purchasing leadership, when requested, the supplier's quality data as needed for supplier executive reviews.
SQE/GCL	SQ CT Global Champion / regional SQ support - follows the GM 1927 – APQP Task 3 – Sourcing Eligibility.

Reference Documents:

- GM 1927 08 Potential Supplier Assessment (PSA)
- GM 1927 16 Process Control Plan Audit (PCPA)
- GM 1927 71 Quad Report
- GM 1927 78 Step Down Action Plan
- CT Sourceability Reports
- GM 1927 – APQP Task 3 – Sourcing Eligibility

Task Number: 6

Task Name: Global Emerging Issue/Alert Process

Task Owner: SQ Plant Support

Task Timing: Throughout the life of the product

Task Description: The purpose of the Global Emerging Issue/Alert Process is to implement standardized work for GPSC Supplier Quality, GM Quality Organization and the GM supply base, specifically focused around communication and initial activities for emerging quality issues that may impact Manufacturing or Assembly Centers, or any other receiving locations in the same or other regions. The intent is to minimize potential impact of emerging quality issues through effective communication, containment activities, and established break points with certified product.

Criteria for Application:

- Part Quality Concerns that potentially impact the following areas:
 - 1) Quality Gates.
 - 2) Global Customer Audit (GCA)
 - 3) Direct Run Rate, Direct Run Loss, FTQ
 - 4) C.A.R.E
 - 5) Vehicle repair float
 - 6) Excessive repair time
 - 7) Main assembly line downtime
 - 8) Vehicle Shipping yard on 'HOLD' status
- 24hr CDP (Concern Detection Process)

Key Deliverables:

- Global Emerging Issue Alert (GM 1927 65)
- Conference call Schedule
- Certified Material/Break Points
- Controlled Shipping (if required)

Customer for Deliverables: GM Manufacturing, Assembly centers/SPO/other receiving locations, Supplier Quality.

Necessary Inputs:

Problem description
Global Customer Audit (GCA)
Direct Run Rate, Direct Run Loss, verification stations
C.A.R.E.
Shipping Hold
Vehicle Repair Float
Main Line Downtime

Source of Input:

Assembly Center/SQ /QE
Assembly Center
Assembly Center
Assembly Center
Assembly Center
Assembly Center
Assembly Center

Resources: Supplier Management, Supplier Quality Engineer/Supervisor, Quality Engineering, 3rd Party Provider (if applicable), PC&L/MSO (if applicable),

<u>Responsible</u>	<u>Methodology:</u>
	Assessment Phase
QE/SQ	A significant quality concern is identified at a GM manufacturing location which has potential for meeting the one or more criteria described in the global major disruption definition.
QE/SQ	Determine potential severity and identify appropriate stakeholders/resources.
QE/SQ	Immediately communicate concern to local leadership and SQ Plant Support Manager
QE/SQ	Determine if other GM locations are affected and notify Quality Leadership at each location
QE/SQ	Assesses which of the following situations exists: Situation A: Supplier is located in the same region as all GM receiving locations Situation B: Supplier is located in the same region as the detecting GM location, but ships to an additional GM location in another region.

	Situation C: Supplier is located in different region from the detecting location
QE/SQ	Issue SPPS to supplier - Reference GM 1927 17 Supplier Quality Processes and Measurements Procedure
	Implementation Phase
QE/SQ	Situation A; Contact the supplier and immediately execute the following: Complete/distribute the Emerging Issue Report: - Translate issue into print or boundary specifications. - Establish sort criteria. - Understand potential root cause. - Understand scope of problem (when did it potentially start). - Method to certify material. - Establish break point and method to identify certified material. - Contain and certify the pipeline. - Notify receiving locations of certification ID and estimated time of arrival of certified material.
QE/SQ	Situation B, Follow activities for Situation A and also send the Emerging Issue Report to the single point contact in the affected region (reference Cross Regional Support Matrix).
QE/SQ	Situation C, Follow activities for Situation A and also do the following: Send the Emerging Issue Report to the single point contact in the region responsible for that supplier (reference Cross Regional Support Matrix.) Set up a conference call with the supplier and the responsible SQE representative to execute the steps referenced in Situation A.
QE/SQ	Set up subsequent conference calls to complete activities.
	Monitor and Check Phase
QE/SQ	Clearly understand the roles and responsibilities for subsequent tasks.
QE/SQ	Subsequent activities outside the scope of this task include: -Verify the break point is still intact - Drill Deep Wide Worksheet analysis GM 1927 84 -Verification of root cause -Corrective Action Additional process controls (error proofing, layered auditing, etc.)
Regional SQ	Travel to Supplier's Location to confirm containment and corrective actions
Supplier	Communicate the Emerging issue to all affected Customer Locations. Execute activities specified in "Situation A". Drill Deep Wide Worksheet analysis GM 1927 84 Timely containment and response to SPPS per GM 1927 17 Supplier Quality Processes and Measurements Procedure Attend all Conference calls as required. Participate in any/all follow up reviews/audits at the Supplying location.

Reference Documents:

GM 1927 17 Supplier Quality Processes and Measurements Procedure

GM 1927 65 Global Emerging Issue Alert

GM 1927 84 Drill Deep Wide Worksheet analysis

Global Emerging Issue Definition

***Completion of the various activities in this task may require the collaboration of a number of
It is critical that these QE's/ SQE's understand their roles and deliverables.*

QE's /SQEs.

Task Number: 7

7

Task Name: Controlled Shipping Level 1 (CS1)

Task Owner: Supplier Quality Engineer (SQE)

Task Timing: Current Production, response to Supplier quality concerns

Task Description: Controlled Shipping Level 1 (CS1) is a demand by Customer that a supplier put in place a redundant inspection process at the supplying location to sort for a specific and specified nonconformance, implement a root-cause and problem solving process, and isolate Customer from the receipt of nonconforming parts/material. The redundant inspection is in addition to normal controls, is enacted by the supplier and must be in addition to the normal production process controls. Suppliers that provide products to multi-regions will require that the GM SQ in the supplier region provide global lead during their regional supplier's CS1 process.

Additional Description: The data obtained from the redundant inspection process is critical as both a measure of the effectiveness of the secondary inspection process and the corrective actions taken to eliminate the initial nonconformance. Recurrence for the same failure mode at the Customer Manufacturing Centers during the CS 1 process is another measure of the effectiveness of the corrective actions and containment process.

Criteria for Application (any of the items below may be considered):

- Repeat SPPS's
- Supplier's current controls are not sufficient to ensure conformance to requirements
- Duration, quantity, and/or severity of the problem
- Internal/External Supplier data
- Quality Problem in the field (i.e. PRTS, Warranty, JD Power)
- Major Disruptions

Key Deliverables:

- Certified Material to the Customer facility
- Problem solving capability to establish root cause
- GM 1927 66 CS1 and 2 I Chart to demonstrate product quality **conformance** (nonconforming vs. quantity checked)
- Documented and verified Corrective Action
- Additional process controls to prevent future occurrences
- Revised PCPs, PFMEA, Standardized work instructions as required
- GM 1927 54 CS1 / CS2 Exit Checksheet
- Controlled Shipping Level 1 Entry letter sent from SPPS
- Corrective Action Report from supplier
- CS1/ CS2 Exit Letter sent from SPPS.
- Approved DDW and DDW Read Across (GM 1927 84 and GM 1927 69)

Customer for Deliverables: Customer Assembly/Manufacturing Centers, SQE

Necessary Inputs:

Supplier Practical Problem Solving (SPPS)
Request for Entry into CS1

Source of Input:

Mfg/Assembly Centers/SQE / SPPS
Mfg/Assembly Centers/SQE / SPPS

Resources: Customer Plant Quality/Reliability Department, Customer Supplier Quality Engineer, Buyer, and Supplier

Responsible	Methodology:
Manufacturing Center / SQE	Assessment Phase Manufacturing Center/SQE make a request for CS1, referencing the non-conformances identified either at the Customer Plant or at the supplier facilities. SQE reviews the request to ensure it complies with the criteria for application, and if applicable, makes the decision to place the supplier in CS1. This decision may also involve a SQ Manager.
SQE	Entry/ Implementation Phase Customer Issue Owner (CIO) verbally notifies the supplier they are being placed in CS1 and that a confirmation letter will be sent from SPPS. CIO enters the CS1 record into SPPS to allow the system to deliver the entry letter to the supplier automatically.

Reference Documents:

- GM 1927 17 Supplier Quality Processes and Measurements Procedure
- GM 1927 54 CS1/CS2 Exit Checksheet
- GM 1927 66 CS1 and 2 I-Chart
- GM 1927 69 Drill Wide Matrix
- GM 1927 84 Drill Deep Worksheet
- GM 1927 85 Drill Deep and Wide Audit

SQE in this document refers to the SQE or other approved Customer Representative

SQ Manager in this document refers to a SQ Manager or other Customer Manager

Task Number: 8

Task Name: Controlled Shipping Level 2 (CS 2)

Task Owner: Supplier Quality Engineer (SQE)

Task Timing: Current Production, response to Supplier Quality concerns

8

Task Description: Controlled Shipping Level 2 is a Customer requirement to a supplier to put in place a 3rd Party redundant inspection process to sort for a specific nonconformance, and implementing a root-cause problem solving process. The redundant inspection is in addition to normal controls and Controlled Shipping Level 1. Suppliers that provide products to multi-regions will require that the GM SQ in the supplier region provide global lead during their regional supplier's CS2 process.

Additional Description: The 3rd party or a Customer representative will perform assessment audits. The data obtained from the 3rd party redundant inspection process as well as the audits are critical as both are a measure of the effectiveness of the secondary inspection process and the corrective actions taken to eliminate the initial nonconformance.

Criteria for Application (any of the items below may be considered):

- Repeat SPPSs
- Supplier's current controls are not sufficient to ensure conformance to requirements
- Duration, quantity, and/or severity of the problem
- Internal/External Supplier data
- Quality Problem in the field (i.e. PRTS, Warranty, JD Power)
- Major Disruptions
- Controlled Shipping Level 1 failures

Key Deliverables:

- Certified Material to Assembly Plant
- I-Chart (GM 1927 66) to demonstrate product quality conformance (nonconforming vs. quantity checked by a 3rd party)
- Problem solving capability to establish root cause
 - Correct and effective root cause analysis (Technical, Detection, Systemic)
 - Documented and verified irreversible corrective actions and additional controls to prevent future recurrences
 - Revised PCP's, PMEA, Operator Instructions as required
- Approved DDW and Read Across (GM 1927 84 and GM 1927 69)
- Supplier notifies their IATF 16949 Registrar of CS2 within 5 days of entry

Customer for Deliverables: Customer Assembly/Manufacturing Centers, SQE

Necessary Inputs

Supplier Practical Problem Solving (SPPS)
Request for entry into CS2 through SPPS System

Source of Input:

Mfg/Assembly Centers/SQE / SPPS
Mfg/Assembly Centers/SQE / SPPS

Resources: Customer Plant Quality/Reliability Department, Customer Plant Supplier Quality Engineer, Buyer, Tier1 supplier, and 3rd Party Provider

<u>Responsible</u>	<u>Methodology:</u>
SQE/QE	<u>Assessment Phase</u> Manufacturing Center/SQE make a request for CS2, referencing the non-conformances identified either at the Customer Plant or at the supplier facilities. SQE reviews the request to ensure it complies with the criteria for application, and if applicable, makes the decision to place the supplier in Controlled Shipping. This decision may also involve a SQ Manager.

SQE	<u>Entry/ Implementation Phase</u> SQE verbally notifies the supplier they are being placed in CS2, that an SPPS system generated confirmation letter will follow and that they must contract a GM approved 3rd party CS Provider (Posted in GM SupplyPower – Quality Library – 3rd Party Provider Management). SQE enters the CS2 record into SPPS in the Containment section to allow the system to deliver the entry letter to the supplier automatically. • Supplier selects and contacts a Controlled Shipping 3rd party (provider) and issues a purchase order for Controlled Shipping Level 2 activities within 24 hours of receiving the CS2 letter. • Supplier returns confirmation letter to the SQE SQE conduct the CS 2 kick off meeting (via conference call or on site meeting) with the Supplier’s Quality Manager and Plant Manager to: Review the non-conformance that resulted in the CS2 entry. - Review and approve the supplier’s containment process which includes: ▪ Control of non-conforming product. ▪ Lay out and inspection equipment - Review and approve the CS 2 Third Party’s containment process which includes: ▪ Data collection utilizing an I-chart. ▪ Communication back to Customer (including frequency). ▪ Inspection instructions ▪ Material identification after inspection - Review and approve the supplier’s escalation/reaction plan for the containment activity. - Establish boundary samples (if applicable) and/or specifications for acceptance/rejection of the parts. - Establish exit criteria for the CS2. Default criteria as follows: ▪ 20 consecutive working days of data (from the date of implementation of permanent corrective action) which verifies that the normal production controls are effective for controlling the discrepancies identified in the Controlled Shipping activity. ▪ Documentation showing the root cause was identified and verified. ▪ Documentation indicating that corrective action was implemented and validated. ▪ Copies of all documentation revised as required (Control plan, FMEA’s, Process Flow, operator instructions, etc.) ▪ Documentation indicating that every effort was taken to implement error proofing. ▪ Documented evidence DDW/Read across completed to close SPPS SQE requests the support of the Customer buyer if the supplier is uncooperative in implementing CS2 to Customer’s requirements.
Supplier & CS 2 3rd Party Provider	Implement CS 2 based on the requirements defined in SPPS and SPPS CS2 Entry Letter and confirmed by the SQE during the kick off meeting. Notify additional GM facilities that use the same part, inform them of the nonconformance, and provide containment activities as necessary. Track breakpoints of nonconforming material. (Purge pipeline of suspect material, i.e. at Customer’s facility, in transit and at all storage locations.)
Supplier & CS 3rd Party Provider	<u>Entry/ Implementation Phase</u> Commence the sort activities and display the results in a public and visible location. The 3rd Party Provider performs an additional redundant inspection of all suspect non-conforming products per the agreed upon process to ensure defect free parts are delivered to Customer. If applicable, 3rd party Provider Quality Engineer or Customer SQE reviews the Supplier’s process and quality history, and completes the Assessment. The supplier shall notify its IATF 16949 Registrar within 5 business days after being placed in CS2. Supplier submits effective corrective action plans to the IATF16949 registrar for review and/or assessment and, when closed, Certification Body verifies that the corrective actions have been

	implemented and read across to the entire organization's site. Supplier also authorizes IATF 16949 Registrar to submit the review and/or assessment findings to the Customer.
Supplier	<u>Monitor and Check Phase</u> Perform a redundant inspection of all suspect non-conforming products per the agreed upon process and ensure defect free parts/material are delivered to Customer and document inspection results on the I charts Determine and demonstrate the root cause to the Customer SQE. Develop, implement and validate the permanent corrective actions, along with improved process controls (i.e., error proofing, layered audits, setup checklists, standardized work, operator training and certification program, etc.) and update all the applicable documents like PFMEA, PFD, etc. Conduct a daily management meeting at the sort location to review the results, ensure the corrective actions taken are effective, and plan required changes. Provide the 3rd Party provider the elements required to perform the CS 2 inspection according to the requirements established by the SQE (inspection area, lighting, inspection tables, etc.)
SQE	Monitor supplier's containment data (I-chart.) Verify the supplier has a documented process control to ensure the problem (failure mode) reported by the Customer will never happen again Verify supplier's root cause analysis and corrective actions. Resolve any discrepancy in the interpretation of the inspection criteria between the supplier and the 3rd Party Provider Interface between the supplier and the CS 2 Third Party provider to ensure the CS 2 process is conducted without interruptions.
Supplier	<u>Verification for Exit</u> Meet the defined exit criteria. Request exit from CS 2 and provide supporting documentation and assessments on performance and corrective actions to the appropriate Customer representative (Customer plant representative notified if required). Upon exit of CS2 the supplier should complete the CS2 Exit Survey 3rd Party Provider Services GM 1927 58 and send to the GM Regional 3rd Party Provider representative and upload to SCMS.
SQE	Conduct PCPA Audit, if applicable. Verify that the supplier has met all exit criteria. (Assessment may be performed by the 3rd party.) The Supplier is removed from CS2 after all exit criteria are met and the established time has been completed without further non-conformances at the Customer Assembly /Manufacturing Center or coming out of the Supplier's process. Notify the supplier verbally that they have met the criteria and that they will be removed from CS2 upon exit in SPPS system, which will send a CS2 exit letter. Complete a GM 1927 54 CS1 and 2 exit checksheet (as required) (Distribute exit check sheet to any additional Customer resources who have responsibility for letter creation and distribution, if applicable.) Enter the "Date Out" of CS2 in SPPS. A Controlled Shipping Exit letter will be sent to the supplier automatically.

Reference Documents:

GM 1927 17 Supplier Quality Processes and Measurements Procedure

GM 1927 51 CS2 Assessment Matrix

GM 1927 52 CS Level 1 and 2 Assessment

GM 1927 54 Controlled Shipping Exit Checksheet

GM 1927 58 Controlled Shipping Level 2 Exit Supplier Survey 3rd Party Provider

GM 1927 59 Controlled Shipping 1 2 Kick-off Meeting

GM 1927 66 CS1 and 2 I-Chart

GM 1927 69 Drill Wide Matrix

GM 1927 79 Supplier Corrective Action Plan Form

GM 1927 84 Drill Deep Worksheet

GM 1927 85 Drill Deep & Wide Audit

SQE in this document refers to the SQE, QSE, 3rd Party PQE, or other approved Customer Representative

SQ Manager in this document refers to a SQ Manager or other Customer Manager

Task Number: 9

Task Name: Supplier Performance Review Meetings

Task Owner: GPSC – Current Supplier Quality

Task Timing: Throughout the life of the product

Task Description: Formal meetings between the leadership of the Supplier and the leadership of GM to review the Supplier's quality performance and actions for driving improvement. This task describes 3 types of meetings, each with specific drivers, which engages leadership involvement to improve supplier performance.

Criteria for Application:

- **Supplier Performance Review** – focus is generally a supplier manufacturing Duns performance to a specific GM Assembly/Manufacturing plant. Attendance includes the GM Plant QE and or QE lead, QE and or SQ Manager, the GM Plant Manager and/or Quality Manager, and the Supplier's General/Plant Manager and Quality Manager.
- **Supplier Quality Improvement Meeting** – focus is typically a supplier's ultimate duns performance to GM. Attendance includes the GM Supplier Quality Director(s)/Group Director, GM Supplier Quality Manager(s), the Supplier's CEO/President, Corporate Quality Director, VP Manufacturing.
- **Plant Disruption Review Meetings** – focus is on the assembly plant disruption caused by a supplier manufacturing Duns location. Attendance should include the GM Supplier Quality Director(s), GM Purchasing/Commodity Director and/or Manager, GM buyer, GM Supplier - Quality Manager(s), the Supplier's CEO/President, Corporate Quality Director, VP Manufacturing. Assembly Plant Quality i.e QE, QE Manager, Quality Manager

Key Deliverables:

- Supplier presentation of problem definition, root cause and corrective action taken to protect GM and its customers from failure modes and or quality defects as described in the SPR notification supplied by GM.
- Supplier action plan.
- Completed and Distributed GM 1927 77 Global Read Across Form and Procedure GRA.
- Lessons Learned Database of GRA items stored in Auros.

Customer for Deliverables: GM Assembly/Manufacturing Plant Quality, GM Supplier Quality Engineer/Manager/Director/GM Purchasing Commodity Manager/Director

Necessary Inputs:

Supplier Performance Report (9 panel)
SPPS Analysis Summary
GM Assembly Plant/3rd Party quality data
APQP
Launch
Supplier's Internal / External Data Analysis
Drill Deep/Read Across

Source of Input:

Supply Power
Supply Power / SPPS System
Plant QE/3rd Party
SQE
SQE
Supplier / SQE/QE
Supply Power / Supplier

Resources: Supplier Quality, Purchasing, Supplier, GM Assembly Plant Personnel QE

Responsible	Methodology :
SQE/QE	Pre-Meeting preparation. • Reviews 6-month SPPS history, 9-panel charts, and other reports for targeted supplier to identify trends. • Creates a Pareto for quality non-conformances to identify repeat issues and/or major causes of customer dissatisfaction. • Issue notification letter inviting the supplier to the meeting. (Suggested format) • Prepare a presentation package for review at the meeting with:

	- Meeting agenda (suggested format) - Supplier Performance Report (9-panel) - Supplier's performance metrics to the GM Assembly/Manufacturing Plant(s). Supplier CT Sourceability Report (optional)
SQE/QE	Conduct Meeting. SQE/QE kicks off the meeting by presenting the agenda. The agenda should include: - Safety evacuation procedure - Participant introductions (an attendance list to be completed by all participants) - Intent and expectations
SQE/QE	SQE/QE reviews the Supplier's performance and specific issues as appropriate (9 panel data, GM Plant impact, warranty /field performance)
Supplier	Supplier presents their information, which should detail: - Pareto analysis of their failures – locations, products, modes - Root cause analysis of failures, if applicable - Specific corrective actions to address the failures - Systemic changes to address poor performance - Propose a quality improvement glide-path with sufficiency plans
SQE/QE	For a plant disruption meeting, review the details of the disruption and the impact on the GM plant.
Supplier	For a disruption meeting, review the root cause, the failures in their process, 5 why /read across and their corrective actions.
SQE/QE	Reviews any open cost recoveries, review open items, assignments, owners and follow-up dates, establish date for follow-up meeting, if necessary.
Supplier	Monitor and check. Drive and documents systemic improvements in the organization. Develops implements and tracks an action plan to resolve open issues.
SQE/QE	Tracks progress, reviews improvements, and verifies implementation. Conducts follow-up as required.

Reference Documents:

- GM 1927 44 Supplier Performance Review Invitation Letter
- GM 1927 45 Supplier Performance Review
- GM 1927 77 Global Read Across Form and Procedure
- GM 1927 78 Step Down Action Plan
- GM 1927 79 Supplier Corrective Action Plan Form
- GM 1927 69 Drill Wide Matrix
- GM 1927 84 Drill Deep Worksheet
- GM 1927 85 Drill Deep and Wide Audit
- GM 1927 86 Drill Deep Root Cause/Read Across workbook

**** SQE denotes GM Supplier Quality Engineer or approved GM SQ Representative***

Task Number: 10

Task Name: 3rd Party Provider Process

Task Owner: GPSC – Supplier Quality

Task Timing: On-going / Support GM Assembly Centers

Task Description: The 3rd Party Provider Approval Process is used to maintain qualified and approved 3rd party providers to support and protect the GM Assembly Centers, GM Propulsion and GM Metal Centers sites from receiving products that have been identified as not acceptable and require special inspection, and handling. The 3rd Party Providers are also used to improve supplier quality performance. GM shall conduct purchasing quotes for the services offered by suppliers as the market or need exist. GM will establish a maximum rate for specified services based on the purchasing quotes, Controlled Shipping Level 2. The 3rd Party is hired by the supplier and located at the supplier manufacturing site or another supplier location as agreed upon by the 3rd party provider and or the GM assembly customer.

Key Deliverables:

- Immediate supplier response to GM quality concerns using GM Approved 3rd party providers.
- GM provides listing Approved 3rd Party Providers that supplier can select and enter in contract with.
- Sustained quality containment and quality improvement using GM approved providers to handle Controlled Shipping Level II or specific Approved GM Workshop or GM Audits.
- Reduced cost of services based on services quoted and max. capped rates, hourly rates and or piece price rate charges.

Customer for Deliverables: GM Assembly Center or GM Plant, Supplier Quality,

Necessary Inputs:

Provide list of potential new and current Approved 3rd Party Providers
Provide SOR of the services needed to GM buyer for RFQ
Determine approved list based on price / quality / service / experience
3rd Party Provider Agreement of services (CS1, CS2)
Supplier placed on Controlled Shipping Level 2
Supplier selection of GM Approved 3rd Party Provider CS2
Supplier selection of 3rd Party Provider Contract Established
Evaluation of new and old GM Approved 3rd Party Providers
SPPS's Supplier Practical Problem Solving
Controlled Shipping Level 1 and Level 2

Source of Input:

GPSC SQ / Buyer / Other
SQ Plant Support
GPSC Buyer / SQ
3rd Party Providers / GM Customers / SQ / Buyer
GM Assembly Center / SQ
Supplier
Supplier
SQ Plant Support
SPPS System
SPPS System

Resources: Supplier Quality, 3rd Party Provider, Supplier, Purchasing, GM Assembly Center / Plant Personnel

3rd Party Provider Methodology: (GM Approved CS2 Providers only)

<u>Responsible</u>	<u>Methodology:</u>
SQ Regional	Determines how many 3 rd party providers need to be on the approved list – regional for such services – Controlled Shipping Level 1 and 2 <u>See Quality Statement of Requirements (SOR) for 3rd Party Providers for Controlled Shipping</u>
SQ / GM Buyer (Regional)	Creates and approves SOR (Statement of Requirements) for the 3 rd Party Provider services that can be sent to potential providers.
SQ / GM Buyer (Regional)	Coordinates with the GM buyer to conduct Pre-RFQ, Tech Review and agreement of SOR or scope of service expectations and recommendation for services based quality, service, competitive price, experience and proper resources as requested by the GM supplier base as needed.
Approved 3 rd Party Provider / Supplier	Approved 3 rd Party Providers will understand GM 1927 17 procedure, understanding of Controlled Shipping (CS) Level 1 and 2 process and timeline requirements and the requirements to provide standardize work in a timely manner. Supplier and 3 rd Party Provider will ensure effective corrective action for the SPPS/failure mode that the CS 1 or 2 was initiated. Ensure that other GM problem solving or other problem solving / resolution tools are used (such as DDW, Read Across, Red X.) Providers must have a procedure for the certification of Provider Quality Engineers (PQE) or a PQE Certification process, layered audits shall be conducted by the 3 rd party provider and or GM.

SQ / GM Indirect Buyer (Global / Regional)	GM buyer / SQ will agree on services capped rates that the 3 rd Party Providers can charge the supplier for the services that they are contracted to perform. GM will determine either capped rates or the specific rates for the different services.
SQ Plant support	Overall responsible for the Approved 3 rd Party Provider process using the voice of the customer feedback / supplier history last 12 months, the requirements being met, the evaluation of the services provided to other OEM's and or other customers. Customer surveys and letters from other customers regarding the 3 rd Party Provider customer satisfaction status.
Global Process SQ & Plant Support	Maintains the Global GM Approved 3 rd Party Providers listing and postings in GM SupplyPower Quality Library – 3 rd Party Provider Management – Services. Maintains the Approved CS2 Providers in Supply Power
Supplier	Provide GM SQ and or GM customer the 3 rd Party Provider Customer Satisfaction Survey feedback GM 1927 58a regarding the services provided by the 3 rd party provider. Documents should be attached to SPPS prior to closure.
SQ Plant Support /SQ	Review current Controlled Shipping services being provided, monitor exit timing and cost on-site review of the services being performed. Layered audit of the services provided to the supplier.
GM Assembly Center / SQ SMT / SQ Plant Support	GM Customer could or shall write Program Management SPPS to the Controlled Shipping 2 providers if the non-conforming material inspection has escaped the 3 rd Party Provider containment process (per GM 1927 17). The 3 rd Party provider is subject to re-certification in the event of inspection failure. Reference potential steps to exit criteria a GM Approved 3 rd Party Provider can be done by conducting periodical assessment annually or at a frequency that has been determined by the specific GM region.

Reference Documents:

GM 1927 17 Supplier Quality Processes and Measurements Procedure
Quality Statement of Requirement (SOR) for the 3rd Party Providers for Controlled Shipping
3rd Party Provider Management Reference Manual – Controlled Shipping Level 2
GM 1927 58 CS2 Exit Customer Satisfaction Survey 3rd Party Provider
GM Approved 3rd Party Providers – CS2 (GM SupplyPower – Quality Library)

SQE in this document refers to the SQE, QSE, 3rd Party PQE, or other approved Customer Representative

SQ Manager in this document refers to a SQ Manager or other Customer Manager

Task Number: 11

Task Name: New Business Hold

Task Owner: Supplier Quality

Task Timing: Throughout the life of the products

Task Description: New Business Hold (NBH) is a process to escalate unresolved quality issues within the supplier's organization in order to get them successfully resolved. While on New Business Hold, that DUNS# will no longer be permitted to quote on new business.

Criteria for Application.

(Representative examples. One or more of the following MAY result in New Business Hold):

- Red on Sourceability Report caused by major, multiple, repetitive failure modes
- Unresolved quality issues over an extended period of time.
- Failure to improve performance in a timely manner.
- Unauthorized process or tool change.

Key Deliverables:

- New Business Hold status on Sourceability Report
- New Business Hold record issued in SCMS

Customer for Deliverables: Supplier

Necessary Inputs:

GM 1927 71 Quad Report
Sourceability Report
SPPS Analysis Summary
Warranty data

Source of Input:

SQE / GM Supply Power
SCMS
SQE - SPPS
SQE

Resources: Supplier, Supplier Quality Engineer

Entry into NBH

- SQE creates/updates Quad Report, gathers supporting documentation, and submits to local management for approval. (If a single point SQE is assigned to the supplier, they should initiate the Quad.)
- SQ in the supplier's region provides the Quad Report and supporting documentation to regional SQ Director for review
- Potential NBH will be reviewed at the Global SQ Directors meeting. –
- Executive SQ Director to review and discuss with the VP Purchasing or Commodity Executive Director.
- If NBH action is rejected SQ Director and Purchasing reviews in Global Product Launch Excellence (GPLe) and recommends next steps.
- If NBH action is approved the SQ and Purchasing meets with supplier to review the NBH exit criteria and the SMT Manager will create the NBH record in SCMS
- [IAOB /IATF may be notified by the SQ Commodity Manager.](#)

Exit from NBH

- SQE monitors supplier's Performance Improvement progress to ensure exit criteria are being met.
- When the supplier has met the exit criteria, the SQE updates the Quad Report with supporting documentation.

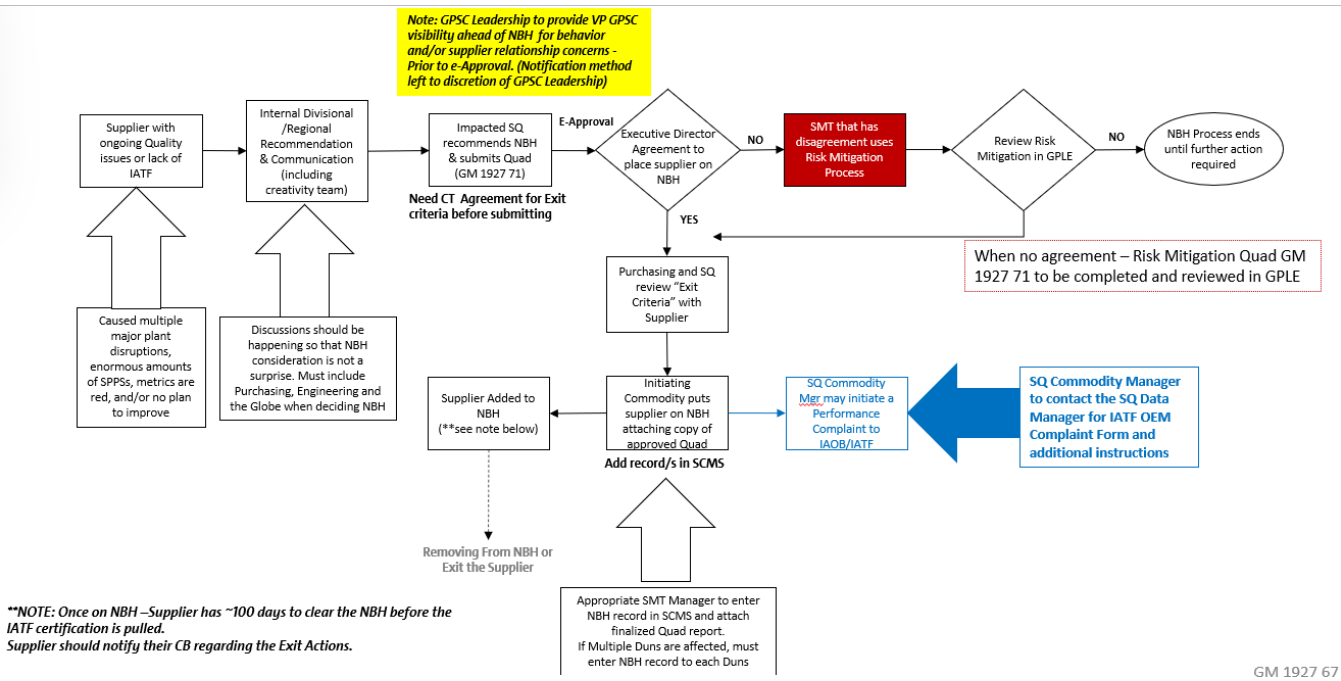
- SQE and SMT manager initiate a meeting with GPSC Commodity Manager and/or Creativity Team to recommend them that the New Business Hold be removed.
- If NBH removal is approved, the SQE has to obtain all approvals, and sends all the documents to the SMT Manager.
- Upon approval, the SMT manager creates NBH removal letter and attains appropriate executive signatures, communicates internally, inputs the NBH status into the Supplier Certification Management System (SCMS) and issues an official NBH notification letter to the Supplier.

Responsible	Methodology:
SQE	Creates Quad Report with supporting NBH data.
SQE	Obtain SMT Manager approval for request.
SQE/SMT Manager	Presents to Creativity Team the consideration for NBH and get agreement for exit criteria.
SMT manager	If the recommended NBH is accepted by CT, the action will be presented to Executive Director.
CT and SMT Manager	If SQ Exec. Dir. and Purchasing leadership approve, CT and SMT Manager Conduct Performance review with the Supplier, and review Exit criteria to be removed from NBH
SMT Manager	Creates NBH record in SCMS, and uploads the Quad report in SCMS. SQ Commodity Manager may notify IAQB /IATF.
Supplier	Reviews the NBH Quad Report and Exit criteria.
SQE	Monitors the supplier performance to meeting the defined NBH Exit Criteria on the Quad Report.
SQE	Upon supplier performance improvement and if the supplier has met the NBH Exit Criteria update the Quad Report.
SQE	Upon approval from the commodity sourcing meeting – Must forward the Quad Report to the GPSC SMT Manager for processing.
SMT Manager	Creates NBH Removal letter with required GM signatures, Closes the NBH record in SCMS. Email sent to NBH distribution of the weekly NBH actions taken.
SQE / Supplier	Reviews next updated Sourceability Report for removal of NBH (Coded N= no or not on NBH or Coded Y=yes currently on NBH) for the supplier specific duns code.

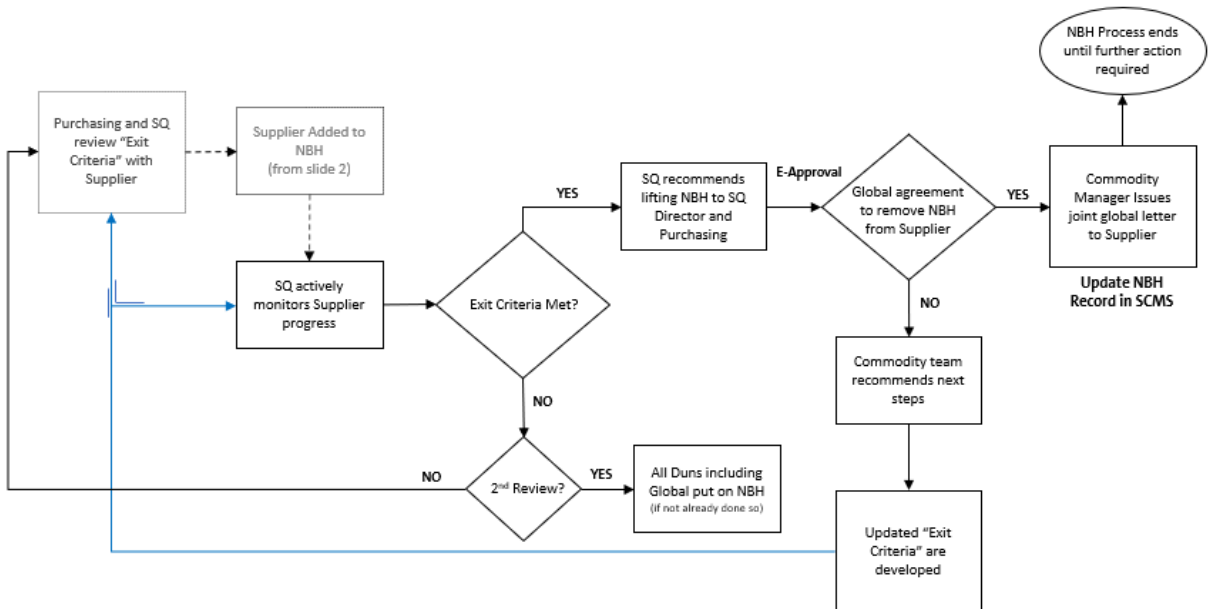
Reference Documents:

- GM 1927 71 Quad Report
- GM 1927 67 NBH process flow chart
- Supplier Certification Management System (SCMS), Sourceability Report and SPPS (Supplier Practical Problem Solving SPPS)
- GM 1927 36 Quality Management Training (5 modules)

New Business Hold Process Flow



Removing New Business Hold Process



Revision History

Global APQP Process

January, 2004

- Task 1 *Commodity Sourcing Strategy Meeting*—Under "Necessary Inputs" added warranty. Under "Methodology" section added types of warranty data to discuss at meeting. Under "Methodology" section added more information on PSA requirements and scoring method. Under "SQE Responsibility" section added that Part Specific SOR needs to be included in RFQ. Under "Additional Information" added AIAG QSA and Data sources for Warranty.
- Task 2 *Technical Reviews*—"Under Necessary Inputs" eliminated Complex Systems/Subassemblies. Under "Methodology" section added Part Specific SOR to lessons learned review. Under the "SQE Responsibility" section added supplier understanding of Part Specific SOR. Under the "SQE Responsibility" section removed the word "recommended" that was listed next to the supplier understanding of SQ SOR. Under "Supplier Responsibility" section replaced "IPTV" with "warranty goals and expectations" and added warranty sufficiency plans.
- Task 3 *GPSC APQP Assessment & Sourcing*—Under "Necessary Inputs" removed "QSA" reference to eliminate confusion (PSA and QSA are the same document). Under the "SQE Responsibility" section modified the follow-up requirements to correspond with changes to the GPSC APQP Assessment changes.
- Task 4 *Supplier Gate Reviews*—Under "Deliverables" section removed sentence that referenced the Complex Systems appendix. Under "Methodology" section removed sentence from each Gate Review that referenced APQP Program management Complex Systems. Replaced "QWIK" with "Warranty Reduction Plans". In Supplier Gate Review #3, removed reference to GM 19272 3M. Under "SQE Responsibilities" section added Part Specific SOR to lessons learned requirement.
- Task 6 *Feasibility and Manufacturing Assessment Letters*—"Methodology" section, added GM Engineer as a recipient of Letters 2-4. Removed sentence that described letter as a formal transfer of responsibility from supplier's engineering to supplier's manufacturing organization.
- Task 8 *DFMEA*—Under "Necessary Input" section replaced "criteria checklist" with "Part Specific SOR". Under the "Methodology" section added "high severity items" to DFMEA flow chart box.
- Task 10 *Gauge, Tooling and Equipment Reviews*—under "SQE Responsibility" section added sentence to verify the existence of GM owned tooling and that it is properly identified as GM property.
- Task 11 *GP-11 Pre-Prototype and Prototype*—Under "Methodology" section added "or equivalent" to requirement that supplier's must ensure subcontractors follow GP-11. Under "Supplier Responsibility" section replaced "GM 1826 2" with "GP-11 Corrective Action Plan"
- Task 12 *PFMEA*—Under "Necessary Inputs" section added Part Specific SOR to lessons learned input. Under "Methodology" section added "high severity items" to PFMEA box. Added "customer complaints, and internal data for the life of the part" to box that describes revision drivers. Under "SQE Responsibility" section added high severity failure modes to action plan requirement. Under "Supplier Responsibility" section add requirement for formal documented RPL reduction process, activity through the life of the part, subcontractor's involvement, and periodic reviews.
- Task 13 *Control Plans*—Changed "may" to "should" for use of AIAG Control Plan format in the Definition section. Under the "Supplier Responsibility" section added requirement for layered audits and daily validation of error proofing. Also added review of subcontractor control plans.
- Task 14 *Early Production Containment (GM 1927 28)*—Under "Definition" section added Early Production Containment applicability whenever mandated by GM on any parts that present significant risk to a customer plant. Under "Methodology" section add requirement for separate inspection area, irreversible corrective action, and 100% check.

- Task 16 *Run @ Rate (GM 1927 35)*—Under "Necessary Inputs" changed "Quoted" to "Contracted" tooling capacity. Under "Methodology" section added that SQE may decide to participate in sub-component Run @ Rate, as appropriate. Under "SQE Responsibility" changed "regional database" to "system" in the bullet referencing where to status customer-monitored Run @ Rates. Under "Supplier Responsibility" section removed "supplier monitored" from bullet referencing the status of Run @ Rates. Under "Additional Information" section removed wording regarding 2001 edition of GM 1927 35. Under "Methodology", "SQE Responsibility", and "Supplier Responsibility" sections added a reference to GM 1960-C1, C2, C3.
- Task 17 *Lessons Learned*—Under "Deliverables" added Part Specific SQ SOR updates. Under "Methodology" section changed "Key Stakeholders" to "Commodity Sourcing Strategy" meeting. Added references to Part Specific SQ run's. Under "Supplier Responsibility" section moved error-proofing validation to Task 13.

Appendices

Appendix 1	Complex Systems/Sub-assemblies APQP Management Plan	Moved Document Usage Guidelines to the beginning of the APQP manual and made Complex Systems Appendix 1.
Appendix 2	Data Sources for Warranty	Moved Complex Systems/Sub-assemblies APQP Management Plan to Appendix 1 and made Data Sources for Warranty Appendix 2.

May, 2005

Appendix 2 – Data Sources for Warranty

Legend Legend updated to correctly show Internal and External versus GM only sources.

June, 2007

Various *Changed references to QS900 and Quality Characteristics to match current terminology.*

September, 2008

- Task 1 *Commodity Key Stakeholders Meeting*—Task Name changed from Commodity Sourcing Strategy Meeting to *Commodity Key Stakeholders Meeting*; Task Description, Key Deliverables, Necessary Inputs and Methodology changed to meet GVDP 5.0 and Engineering & Advanced Purchasing Sourcing Process
- Task 2 *Technical Reviews*— Task Description, Key Deliverables, Necessary Inputs and Methodology changed to meet GVDP 5.0 and Engineering & Advanced Purchasing Sourcing Process; Task 3 was integrated to this Task
- Task 3 GPSC APQP Assessment & Sourcing was removed and integrated to Task 2; SQ NOD 004 was incorporated into the manual at this Task
- Task 4 *Gate Reviews*— Task Name changed from Supplier Gate Reviews to Gate Reviews. Task Time, Description, Key Deliverables, Necessary Inputs and Methodology changed to meet GVDP 5.0 and Engineering & Advanced Purchasing Sourcing Process; PPAP and Run @ Rate Reviews were introduced; The APQP Supplier Status Workbook GM 1927 34 was introduced to support Gate Reviews execution
- Task 6 *Feasibility and Manufacturing Assessment Letters* —Was removed
QSB —Was transferred from Current Section to APQP Section and updated
- Task 7 *Process Flow Charts* — Task Description, Key Deliverables, Necessary Inputs and Methodology changed to meet GVDP 5.0
- Task 8 *DFMEA*— Task Description, Key Deliverables, Necessary Inputs and Methodology changed to meet GVDP 5.0 and introduction of DRBFM activity
- Task 10 *Tooling and Equipment Reviews* — Task Name changed from Gauge, Tooling and Equipment Reviews to Tooling and Equipment Reviews; Gauge activity was removed and transferred to Task 11; Task Description, Key Deliverables, Necessary Inputs and Methodology changed to meet GVDP 5.0

- Task 11 *GP-11 Pre-Prototype and Prototype* — Was Removed.
Gauge Development And Approval — Was Removed from Task 10 and transferred to this Task in order to introduce the complete methodology to support the application of GM1925 (Checking Fixture Standard for purchased parts).
- Task 12 *PFMEA* — Task Description, Key Deliverables, Necessary Inputs and Methodology changed to meet GVDP 5.0
- Task 13 *Control Plans* — Task Description, Key Deliverables, Necessary Inputs and Methodology changed to meet GVDP 5.0 and Powertrain requirements
- Task 14 *Early Production Containment GM 1927 28* — Task Description, Key Deliverables, Necessary Inputs and Methodology changed to meet GVDP 5.0
- Task 16 *Run @ Rate GM 1927 35* — Complete methodology update
- Task 17 *Lessons Learned* — Complete methodology update

Appendices

Appendix 1	Complex Systems/Sub-assemblies APQP Management Plan	Removed
Appendix 2	Data Sources for Warranty	Removed

February, 2009

- Various - Global RASIC included as Global APQP section
- Various – Added Global PFMEA Checklist to the Document Usage Guidelines
- Task 4 *Gate Reviews* – Added clarification of Direct Buy component SQE roles at System/Assembly Gate Reviews
 - Changed Key deliverable from AIAG A-7 to GM 1927 37
- Task 6 *QSB* - Added Rules for QSB application
- Task 10 *Tooling & Equipment Review* – Added the SQ Tooling Engineer roles where available
- Task 11 *Gauge Development & Approval* – Added complement information regarding detailed gauge timing chart
- Task 12 *PFMEA* - Changed Key deliverable from AIAG A-7 to GM 1927 37
- Task 13 *Control Plan* – Included the information that functional requirements from SOR & part drawing must be enclosed by PCP

September, 2009

- Section #3 Tasks 1-13
- Section #3 Complete Current Manual re-write - to reflect current work practices and global process agreements made during Q3 2009 Current Revision supported by the Global Task Team and Global SQ Process team as supported by GMNA, LAAM, GME, GMAP and GMPT.
- Task 11 / Contents Removed obsolete process Executive Champion Process (ECP) and replaced with 3rd Party Provider Process, Task 11 and Table of Contents page IV. Removed word “Spill” replaced with “Disruption” in the contents page IV, document guidelines listing.
- Updated Glossary of Terms pages 93 – 97.

Revised Document Usage Guidelines – Page 48.

Obsolete Documents includes the following:

GM 1927-60, GM 1927-63, GM 1927-64, GM 1927-68, GM 1927-70, GM 1927-74, GM 1927-77, GM 1927-81,

August, 2011

Section #2 Tasks 1-13

- Section #2 *Launch Manual Update* – various minor changes to reflect current work practices.

November, 2011

Section #2 Tasks 4

- Launch Manual Update* – Added reference to GM 1927 15.

June, 2012

Section #1 Tasks 14

- Add optional GM 1927 39 Early Production Containment Exit Letter.*

September, 2013

Section #2

- Removed Launch from the manual and created separate GM 1927 Launch Manual document.

December, 2013

Section #4 Task 4

Changes section from Supplier Gate Review process to Valve Assessments.

Various Sections and Tasks

Changed the balance of document to reference Valve Assessments and the associated timing

April, 2014

Section #4 Task 4

Changes section from Valve Assessments to Supplier Readiness Valves with associated timing.

Various Sections and Task

Changed the balance of document to reference Supplier Readiness Valves (SRV) and the associated timing

January, 2015

Various Sections

Removed the Document Usage Guidelines

Removed older references to GM 1927 23 Directed Buy Checklist

January, 2016

Various Sections

Replaced QSB for BIQS

Replaced GQTS for SQMS

Replaced GM1411 for Action Plan in SQMS

Updated "Document Usage Guidelines" Chart to remove obsolete GM 1927 documents

June, 2017

Task 3 Changed the business process from CTBL – creativity team bidlist into Sourceability Report. This change also apply throughout the document.

Task 6 Updated the BIQ elements to include element #30, 30: BIQ Audit

Task 8/12 Updated all applicable areas to reflect changes from RPN reduction to PFMEA Gap/Risk Analysis.

Section #2 (Global Current) Updated all applicable areas to reflect change from RPN reduction to PFMEA Gap/Risk Analysis.

April, 2018

Various Sections

Updated Greenfield/Brownfield document name (Greenfield/Brownfield Assessment GM 1927 31) and added references as appropriate throughout

Added new document PDCA Program Status GM 1927 031a a document reference as appropriate throughout

Removed all references to GP8; process was replaced by BIQS

Removed all reference to GP 12; Type O corrections – process is and has been Early Production Containment GM 1927 28 and supporting documents

Changed term BIQS Audit to BIQS Assessment throughout the document as applicable

Global APQP Basic

Converted 3rd Matrix in the section from the picture to an editable matrix to allow updating of references to Process and Systems as well as repair numerous type o

Task 16

Replaced GP9 with GM 1927 35 in one location

Glossary of Terms

Added SCMS

July, 2019

Replaced IATF 16949 for BIQS

Replaced BIQS with BIQS Sourceability

Removed dashed from documents names to align to GM standard and SupplyPower

Removed reference to GP 5 ; GP 8; GP 9 and GP 12 and replaced with GM 1927 document no. throughout the document.

Added IATF and SPPS to Glossary of Terms

Task 8; Task 12; Task 13; Task 17 Updated verbiage to include Quality Chain requirements /PFMEA updates

Task 15 added "PDT should agree on Action Plans before submittal."

January 2020 rev. 23

Page V: Deleted GP – General Procedure reference

Various Sections:

The following GP were replaced with updated names of the documents.

- GP 5 = GM 1927 17 Supplier Quality Processes and Measurement Procedure

- GP 8 = Continuous Improvement included in SPPS
- GP 9 = GM 1927 35 Run at Rate
- GP 12 = GM 1927 28 Early Production Containment

Task 6: renamed from BIQS Sourceability to **GM Quality Performance Requirements**

Task 5: Replaced: “detailed review of all timing charts and concerns is conducted periodically” with “ Is conducted at least every 2 months”

Task 12: Replaced: “Starting at Kick-Off Meeting and complete by PPAP” with “Starting with preliminary PFMEA evaluation at Technical Review. Complete by PPAP”

Section #2 (Global Current)

Document Usage Guidelines by Global Region: BIQS Sourceability changed to GM Quality Performance Requirements

- **Task 1:** Updated documents references.
- **Task 2:** Title changed to **GM Quality Performance Requirements**. Updated based on current processes.
- **Task 5:** Updated based on current processes. BIQS assessment (replaced with IATF 16949 requirements) and Top Focus requirement removed.
- **Task 7:** Major Disruption Prevention Activity – Deleted. Task 7 is now CS1
- **Task 12** Top Focus: Deleted
- **Task 11: New Business Hold – new process**

March 2020 rev 24.0

Added reference to “AIAG VDA FMEA Handbook” instead of AIAG FMEA Manual throughout the document to align with new AIAG VDA FMEA requirements

March 2020 rev 25.0

Task 16: “Run at Rate” was updated with additional Build at Rate information. The Task 16 name is now: Run at Rate / Seat Complete (SC) Build at Rate

Task 17: “Lessons learned” was updated to include the new Global Read Across process.

Removed the AIAG VDA FMEA Handbook reference and went back to AIAG FMEA 4th edition.

March 2020 rev 26.0

Task 15: Production Part Approval Process (PPAP). Added Definitions at page 31.

June 2020 rev. 27.0

Task 14: Early Production Containment (EPC) :

Added “until 90 days after SORP and meets exit criteria defined per GM 1927 28” to Task timing.

Added “or after shut down for extended period of downtime” to Task description.

Added: • Pre-Launch Control Plan and • Early Containment Exit Letter GM 1927 39 to Key deliverables.

Updated Methodology.

Added: GM 1927 17 Supplier Quality Processes and Measurements Procedure

- GM 1927 11 APQP Kickoff and SRV Worksheet:
- CG4355 GM 1927 03 SQ Sub Tier Supplier Management Statement of Requirements to Reference Documents.

February 2021 rev 28

Throughout the document:

- GM 1925 GM Fixture Standard changed to **GM 1927 10 GM Fixture Standard**.
- GM 1927 29 Gauge Request Certification changed to **GM 1927 29 Gauge Concept Design and Build review workbook**
- GM 1927 20 Process Capability changed to **GM 1927 20 Measurement System Analysis (MSA) Result Sheet**
- Updated the documents name to match Supply Power.
- Entire Manual revised - to reflect current work practices and current global processes and systems.

Section 1:

Task 4 APQP Kick-off Meeting and Supplier Readiness Valves updated.

Task 6: Run at Rate: Entire Task updated including R@R Methodology.

Task 10: Updated the file name to GM 1927 20 Process Capability Measurement System Analysis (MSA) Result Sheet

Task 11: Update the file name to GM 1927 29 Gauge Concept Design and Build review workbook

Added to Methodology: Supplier: Deliver AIAG MSA studies (BIAS, Linearity, Stability, GR&R and Fleiss Kappa, if applicable) for all measurement devices and add to the PPAP package. Reference form GM 1927 20 Process Capability Measurement System Analysis (MSA) Result Sheet or similar.

Throughout the document: Replaced Powertrain with GM Propulsion (GPS)

Task 16: Run at Rate (R@R) / Seat Complete (SC) Build at Rate (B@R) Updated all including the Methodology Chart /Process Flow.

Section 2.

Deleted the follow documents that are now covered by the SPPS System:

Controlled Shipping Confirmation Reply Form	GM 1927 53
Controlled Shipping Level 1 (CS1) Entry letter	GM 1927 55
Controlled Shipping Level 1 and 2 (CS1 and CS2) Exit Letter	GM 1927 56
Controlled Shipping Level 2 (CS2) Entry Letter	GM 1927 57

Task 6: Added 3rd Party Certification and SPPS dispute resolution to Key Deliverables and Program Team, Quality Operations to Resources.

Task 7 and 8 Updated. Eliminated the following documents that are now covered by the SPPS System

~~GM 1927 55 – Controlled Shipping Level 1 Entry letter~~

~~GM 1927 57 – Controlled Shipping Level 2 Entry letter~~

~~GM 1927 53 – Controlled Shipping Confirmation Reply Form~~

~~GM 1927 56 Controlled Shipping Level 1/2 Exit Letter~~

Task 9: Supplier Performance Review Meetings: Key Deliverables aligned with Read Across Process.

Updated Glossary of Terms

March 25 2021 rev 28.1

Section 2 Task 3: Added clarification regarding Consignment / Bailment suppliers and Shutdown/startup timing.

October 7 2021 rev 28.2

Section 2 Task 7: Page 58: Deleted reference to GM 1927 55 which was deleted (it is now covered by SPPS system)

Section 2 Task 8 and 10: Updated document name. Changed from GM 1927 58a to GM 1927 58 Controlled Shipping Level 2 (CS2) Exit Survey (Supplier)

December 06 2021 rev 28.3

Section 2 Task 11 : Page 69: Updated New Business Hold Process

Throughout the document: Replaced GM 1927 30 BIQS Self assessment with GM 1927 30 Quality Management Gap assessment

December 2022 rev 29.0

Global APQP RASIC: replaced SRV with CRV; Deleted GQTS, added Auros.

Throughout the document: Added reference to “**AIAG FMEA VDA Handbook or AIAG PFMEA 4th Edition and CG4338 GM 1927 03 SQ SOR**” instead of AIAG FMEA Manual only, throughout the document to align with new GM FMEA requirements.

Throughout the document: Updated to RPL instead of RPN to align with new AIAG VDA FMEA Handbook and GM Customer Specific Requirements.

Throughout the document replaced GM 1927 36 BIQS Element Training with GM 1927 36 Quality Management (QM) Elements Training and BIQS Sourceability with Sourceability Levels

Deleted: **3rd Party Provider Methodology: (ESEP (GMNA) / QCC GME) and ESEP / QCC services references**

Section 8- Updated Key deliverables: added: “Correct and effective root cause analysis (Technical, Detection, Systemic)”

Section 10- Removed reference to ESEP/QCC and CS1

Page 7: Global APQP RASIC – updated and consolidated columns CT SQE /GCL

Task 4: Updated the Task name to: **APQP Kick Off meeting and Component readiness valves** and updated the content at Page 13: CRV

Task 6: GM Quality Performance Requirements (QPR): Updated Task timing at Page 15

Task 7: Process Flow Charts : Updated Task timing – added “**reviewed at CRV Assessments**”

Task 8: Controlled Shipping level 2 – Updated **Key deliverables** at page 55 and **Entry/ Implementation** Phase at page 56.

Task 10: Tooling and Equipment Reviews: Page 20 – Updated methodology regarding Review timing for Pull Ahead Production Tools. Added DRE/SQE responsibility. Page 20 - Methodology updated: replaced APQP kick off with Tooling kick off meeting.

Task 11: Gauge Development & Approval: Page 22 - Updated Methodology.

Task 13 Control Plan: Page 26 Removed reference to GM Manufacture Confidence Level/Sample Size Table and replaced with: “Sample size and frequency should be determined based on risk and occurrence of the cause and ensures that the frequency adequately protects the customer by ensuring that the product represented by the inspection/test does not reach the customer before the results of these inspections/tests are known”

Task 15: Production Part Approval Process (PPAP): Page 30: Added pull ahead production tool Note.

Updated **Methodology:** deleted form and replaced with “Confirmation email” for CG2503 Barcode Validation;

Added a row with “responsible”: Product Engineering/PPO-SQ: Page 31

Updated **Key Deliverables:** added: “including “pull ahead production tools”

Task 16: Run at Rate – updated SRV to CRV and Task timing: page 33: added CRV assessments

August 2023 rev 30.0

Page IV Updated Labels and Tags - contact information and also added instructions to how to order Labels and Tags through Supply Power portal.

Page 8 Updated Global APQP RASIC

Task 1: Commodity Key Stakeholders Meeting (CKSM): page 9 Added CG4355 GM 1927 03 Sub Tier Supplier Management

Task 4 APQP Kick Off meeting and Component Readiness Valves: page 14 Updated the task owner and the task name; added one line to Methodology:

SQE	SQE conducts the Kick Off meeting offline (optional).
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Task 15 PPAP: page 32 Added the definition for Submission.

Task 11 NBH: page 62 Updated NBH Entry process. Removed the requirement for supplier to notify IATF.

Replaced with: “SQ Commodity Mgr may initiate the Performance Complaint to IAOB/IATF”, to align with IATF 16949 GM CSR.

Task 11 NBH: page 63 updated Methodology.

Task 11 NBH: Page 64 updated process flow (GM 1927 67 NBH Process)

Glossary of Terms

AAR: Appearance Approval Report

ADV: Analysis/Development/Validation

ADV P&R: Analysis/Development/Validation Plan and Report. This form is used to document a Supplier's ADV Plan and track execution

AIAG: Automotive Industries Action Group, an organization formed by General Motors, Ford and Daimler-Chrysler to develop common standards and expectations for automotive suppliers.

AP: Advance Purchasing

APQP: Advanced Product Quality Planning

APQP Project Plan: A one-page summary of the GM APQP process that describes the tasks and the timeframe in which they occur.

AQC: Attribute Quality Characteristic

ASQE: Advanced Supplier Quality Engineer

BIQS: Built in Quality Supply Based

BIW: Body in White. Usually the bare metal shell of the body; includes doors and deck lid prior to paint and trim.

BOM: Bill of Materials

BOP: Bill of Process

Brownfield Site: An expansion of an existing facility.

CKSM: Commodity Key Stakeholders Meeting

CMM: Coordinate Measuring Machine

Cpk: Capability Index for a stable process

CRV: Component Readiness Valve

CTC: Component Timing Chart (DRE document)

CTS: Component Technical Specifications

CVER: Concept Vehicle Engineering Release

DAQ: Design Appearance Quality (Color Harmony / Appearance Group)

Defect outflow detection: A phrase used in the Supplier Quality Statement of Requirements that refers to in-process or subsequent inspection used to detect defects in parts.

DFFA: Dimensional Fit Function & Appearance

DFM/DFA: Design for Manufacturability / Design for Assembly

DFMEA: Design Failure Modes and Effects Analysis. It is used to identify the potential failure modes of a part, associated with the design, and establish a priority system for design improvements.

DPV: Defects per vehicle

DR: Documentation Required

DRE: Design Release Engineer

DSO: Design Sign Off

E&APSP: Engineering & Advance Purchasing Sourcing Process.

Error Occurrence Prevention: A phrase used in the Supplier Quality Statement of Requirements that refers to poke yoke or error-proofing devices used to prevent errors in the manufacturing process from occurring.

ESEP: Enhanced Supplier Engagement Process using GM approved 3rd Party Providers for containment sorting & rework

EWQ: Engineering Work Order

FTQ: First Time Quality

GA: General Assembly

GD&T: Geometric Dimensioning & Tolerancing

GM: General Motors

GMAP: General Motors Asia Pacific

GMIO: General Motors International Operations

GMNA: General Motors North America

GMM: General Motors Mexico

GMSP: GM Supply Power

GPDS: Global Product Description System
GPLE: Global Product Launch Excellence
GPS: Global Purchasing System
GPSC: Global Purchasing & Supplier Chain
GQTS: Global Quality Tracking System
GR&R: Gauge Repeatability and Reproducibility
Greenfield Site: A new supplier facility that is built to support a program.
GSTS: Global Status & Timing System
GVDP: Global Vehicle Development Process
IATF: International Automotive Task Force
IPTV: Incidents per Thousand Vehicles
IVBR: Integration Vehicle Build Release
IVER: Integration Vehicle Engineering Release
KCC: Key Control Characteristics. It is a process characteristic where variation can affect the final part and/or the performance of the part.
KCDS: Key Characteristic Designation System
Kick-Off Meeting: The first APQP supplier program review.
KPC: Key Product Characteristic. It is a product characteristic for which reasonably anticipated variation could significantly affect safety, compliance to governmental regulations, or customer satisfaction.
LCR: Lean Capacity Rate. It is the GM daily capacity requirement.
MCE: Mid Cycle Enhancement
MCR: Maximum Capacity Rate. It is the GM maximum capacity requirement.
MDC: Mule Design Complete
MOP: Make or Purchase
MPC: Material Production Control
MPCE: Material Production Control Europe
MRD: Material Required Date; date material must be delivered in order to allow a build event to begin.
MSA: Measurement Systems Analysis
MU: Mock-Up. Mock-ups can be actual functioning parts, sub-assemblies or assemblies or non-functional wood, expanded polystyrene, plastic or metal representations of parts, subassemblies or assemblies. The design source can be responsible for providing mock-ups.
MVBns: Manufacturing Validation Build non-saleable
MVBs: Manufacturing Validation Build saleable
NBH: New Business Hold
N.O.D.: Notice of Decision
NOA: Notice of Action
OEM: Original Equipment Manufacturer
PAD: Production Assembly Documents
PC&L: Production Control & Logistics
PDT: Product Development Team
PET: Program Execution Team
PFMEA: Process Failure Modes and Effects Analysis.
PPAP: Production Part Approval Process
Ppk: Performance index for a stable process
PPM: 1) Program Purchasing Manager, 2) Parts Per Million (Calculation of discrepant parts /receipts as defined in GM 1927 17 Supplier Quality Processes and Measurements Procedure)
PPV: Product & Process Validation
PQC: Product Quality Characteristic
PQRR: Program Quality Readiness Review
PSA: Potential Supplier Assessment, a subset of the Quality System Assessment (QSA)
QCC: Quality Confirmation Center (GME form of ESEP – enhanced supplier engagement process using GM approved 3rd party providers for containment sorting and rework)
QSA: Quality System Assessment
QRD: Quality Reliability Durability
QTC: Quoted Tool Capacity
RASIC: Responsible, Approve, Support, Inform, Consult
R@R: Run at Rate
RFQ: Request For Quotation
RPL: Risk Priority Level related to FMEA development
RPN: Risk Priority Number
RPL Reduction Plan: An action plan that describes what is being done to reduce the risk priority level for items listed in the DFMEA or PFMEA.
SCMS: Supplier Certification Management System
SDE: Supplier Development Engineer
SFMEA: System Failure Mode and Effects Analysis
SMT: Systems Management Team
SOA: Start of Acceleration
SORP: Start of Regular Production

SOR: Statement of Requirements
SOS: Start of Sales
SPC: Statistical Process Control
SPO: (General Motors) Service and Parts Operations
SPPS: Supplier Practical Problem Solving
SQ: Supplier Quality
SQE: Supplier Quality Engineer
SQIP: Supplier Quality Improvement Process
SQMS: Supplier Quality Management System
SRV: Supplier Readiness Valve
SSF: Start of System Fill
SSTS: Sub-system Technical Specifications
Sub-Assembly / Sub-System: An assembly of sub-components delivered to the GM main production line for installation to the vehicle as a single unit.
Subcontractor: The supplier of a sub-component to a Complex System/Subassembly supplier (Tier 2, 3, etc.).
SVE: Sub-System Validation Engineer
SVER: Structure Vehicle Engineering Release.
Team Feasibility Commitment: An AIAG APQP form that is provided with the Request for Quotation. It is used to document the supplier's concerns with the feasibility of manufacturing the part as specified.
TKO: Tooling Kick-Off
TWO: Temporary Work Order
UG: Unigraphics
VDP: Vehicle Development Process
VLE: Vehicle Line Executive
VLT: Vehicle Line Team
VOC: Voice of the Customer
VOD: Voice of Design
VPI: Vehicle Program Initiation
VTC: Validation Testing Complete
VTs: Vehicle Technical Specification
WO: Engineering Work Order